



```

name: InactivityAndMarriage
log: C:\Users\hansg\Documents\working\SHARE data 5\output.smcl
log type: smcl
opened on: 22 Jun 2026, 13:40:56

```

```

1 .
2 . use "C:\Users\hansg\Documents\working\SHARE data 5\dataset.dta"

3 .
4 . asdoc, text(\par \qc The Influence of Health, Marriage and Cohabitation on Physical
> Inactivity: Evidence from SHARE 2015 to 2022 for 12 European Countries) fs(12) save
> (report.doc) replace
(file report.doc not found)
Click to Open File: report.doc

5 . asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
> chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10) save(report.
> doc) append
Click to Open File: report.doc

6 .
7 . ***variables preparation and data description
8 .
9 . asdoc, text(\page) save(report.doc) append
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10.
11. *panel level id
12. egen id= group(mergeid)

13. drop *_f

14. drop if phinact== -99
(24,644 observations deleted)

15.
16. xtset id Qyear

Panel variable: id (unbalanced)
Time variable: Qyear, 2015 to 2022, but with gaps
Delta: 1 unit

17.
18. asdoc xttrans phinact, freq save(report.doc) append

```

Physical inactivity	Physical inactivity		Total
	0	1	
0	51,749 91.29	4,940 8.71	56,689 100.00
1	2,527 43.80	3,243 56.20	5,770 100.00
Total	54,276 86.90	8,183 13.10	62,459 100.00

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19.

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20. tabulate country phinact

Country identifier	Physical inactivity		Total
	No	Yes	
Austria	6,082	736	6,818
Germany	9,414	838	10,252
Sweden	8,500	476	8,976
Spain	7,904	1,921	9,825
Italy	8,582	3,068	11,650
France	7,733	1,034	8,767
Denmark	8,288	515	8,803
Greece	10,995	1,215	12,210
Switzerland	6,360	327	6,687
Belgium	10,625	1,569	12,194
Czechia	9,139	1,162	10,301
Poland	3,789	862	4,651
Total	97,411	13,723	111,134

21.

22. generate mst=0

23. replace mst=1 if mstat==3 | mstat==5
(11,008 real changes made)

24. replace mst=2 if mstat==4
(6,113 real changes made)

25. replace mst=3 if mstat==6
(17,718 real changes made)

26. label define msL 0 "Officially with partner" 1 "Away from partner" 2 "Never married"
> 3 "Lost partner"

27. label values mst msL

28.

29. spearman mst single

Number of observations = **111,134**
Spearman's rho = **0.8537**

Test of H0: **mst** and **single** are independent
Prob = **0.0000**

30.

31. generate change=0

32. replace change=1 if mst==0
(76,295 real changes made)

33. bysort id: generate changed = change - change[_n-1]
(48,675 missing values generated)

34. replace changed=changed+1
(62,459 real changes made)

35. replace changed=1 if changed==.
(48,675 real changes made)

36. label define chL 0 "No longer officially together" 1 "All other situations" 2 "Newly
> officially together"

37. label values changed chL

38. asdoc tabulate changed, save(report.doc) append

changed	Freq.	Percent	Cum.
No longer officially together	2,072	1.86	1.86
All other situations	108,733	97.84	99.70
Newly officially together	329	0.30	100.00
Total	111,134	100.00	

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39.

40. bysort id: generate schanged=single - single[_n-1]
(48,675 missing values generated)

41. replace schanged=0 if schanged==.
(48,675 real changes made)

42. replace schanged=schanced+1
(111,134 real changes made)

43. label define schL 0 "Hooked up" 1 "Unchanged" 2 "Back single"

44. label values schanged schL

45. asdoc tabulate schanged, save(report.doc) append

schanged	Freq.	Percent	Cum.
Hooked up	430	0.39	0.39
Unchanged	107,804	97.00	97.39
Back single	2,900	2.61	100.00
Total	111,134	100.00	

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46.

47. generate mortg=0

48. replace mortg=1 if mort>0
(17,873 real changes made)

49. label define mlab 0 "No mortgage" 1 "Mortgage"

50. label values mortg mlab

51.

52. codebook phinact sphus thinc mst changed schanged cjs age yedu gender country bmi eu
> rod, compact

Variable	Obs	Unique	Mean	Min	Max	Label
phinact	111134	2	.1234816	0	1	Physical inactivity
sphus	111134	5	3.114771	1	5	Self-perceived health - US scale
thinc	111134	56252	32869.01	0	7399056	Total household income - Version
> A						
mst	111134	4	.6873504	0	3	
changed	111134	3	.9843162	0	2	
schanged	111134	3	1.022225	0	2	
cjs	111134	7	2.24762	-99	97	Current job situation
age	111134	47	70.20186	50	96	Age
yedu	111134	30	11.01304	0	30	Years of education
gender	111134	2	1.561619	1	2	Gender
country	111134	12	18.22554	11	29	Country identifier
bmi	111134	3652	26.82758	12.484	98.61933	Body mass index

eurod	111134	13	2.380928	0	12	EURO depression scale
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53.
54. drop if cjs==99
    (1,089 observations deleted)
55. drop if yedu==9997
    (0 observations deleted)
56.
57. winsor2 thinc, replace cuts(0 99)
58.
59. codebook phinact sphus thinc mst changed schanged cjs age yedu gender country bmi eu
    > rod, compact

```

Variable	Obs	Unique	Mean	Min	Max	Label
phinact	110045	2	.119233	0	1	Physical inactivity
sphus	110045	5	3.106974	1	5	Self-perceived health - US sca
> le thinc	110045	54718	31725.99	0	193799.5	Total household income - Versi
> on A mst	110045	4	.6757599	0	3	
changed	110045	3	.9847608	0	2	
schanged	110045	3	1.021437	0	2	
cjs	110045	6	3.249562	1	97	Current job situation
age	110045	47	70.09174	50	96	Age
yedu	110045	30	11.02087	0	30	Years of education
gender	110045	2	1.560752	1	2	Gender
country	110045	12	18.21905	11	29	Country identifier
bmi	110045	3634	26.8345	12.86854	98.61933	Body mass index
eurod	110045	13	2.371121	0	12	EURO depression scale

```

60.
61. global names thinc
62. foreach num of numlist 1(1)1 {
    2.     local name: word `num' of $names
    3.     bysort country: egen mean`name'=mean(`name') if `name'!=0
    4.     generate T=1
    5.     replace T=0 if `name'==0
    6.     replace T=2 if `name'>mean`name'
    7.     drop mean`name'
    8.     drop `name'
    9.     rename T `name'
    10. }
    (1,712 missing values generated)
    (1,712 real changes made)
    (41,972 real changes made)

```

```

63.

```

```
64. label define thL 0 "No income" 1 "Below country average income" 2 "Above country ave
> rage income"
```

```
65. label values thinc thL
```

```
66.
```

```
67. keep gali sphus thinc mst mstat changed schanged cjs age yedu gender country single
> id Qyear my_wgt phinact bmi eurod
```

```
68.
```

```
69. asdoc spearman phinact gali sphus thinc single mst changed schanged cjs age yedu gen
> der country bmi eurod, save(report.doc) append
```

Number of observations = **110,045**

>	cjs	phinact age	gali yedu	sphus gender	thinc country	single bmi	mst eurod	changed	schanged
	phinact	1.0000							
	gali	0.2505	1.0000						
	sphus	0.3167	0.5409	1.0000					
	thinc	-0.1074	-0.1097	-0.1636	1.0000				
	single	0.0935	0.1006	0.1080	-0.3634	1.0000			
	mst	0.0864	0.0951	0.0975	-0.3040	0.8545	1.0000		
	changed	-0.0192	-0.0224	-0.0292	0.0756	-0.1969	-0.2288	1.0000	
	schanged	0.0254	0.0291	0.0354	-0.0868	0.2461	0.1517	-0.6753	1.0000
>	cjs	0.0027	-0.0832	-0.0675	-0.0037	-0.0483	-0.0443	0.0171	-0.0193
>	1.0000								
>	age	0.2057	0.2143	0.2573	-0.1906	0.2087	0.1991	-0.0822	0.0850
>	-0.4774	1.0000							
>	yedu	-0.1568	-0.0986	-0.2218	0.2222	-0.0712	-0.0738	0.0308	-0.0257
>	-0.0154	-0.2295	1.0000						
>	gender	0.0370	0.0405	0.0266	-0.1028	0.2013	0.1924	-0.0399	0.0371
>	0.1782	-0.0271	-0.0727	1.0000					
>	country	-0.0020	-0.0107	-0.0003	-0.0156	0.0268	0.0297	0.0037	0.0024
>	-0.0193	-0.0507	0.0608	0.0165	1.0000				
>	bmi	0.0517	0.1060	0.1490	-0.0380	-0.0232	-0.0171	0.0088	-0.0095
>	-0.0095	-0.0249	-0.1011	-0.0908	0.0516	1.0000			
>	eurod	0.2090	0.3307	0.4028	-0.1200	0.1436	0.1332	-0.0672	0.0744
>	0.0360	0.1102	-0.1129	0.1822	0.0121	0.0272	1.0000		

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```
70.
```

```
71. label copy cjs cjsLab
```

```
72. label list cjsLab
```

```
cjsLab:
```

```
-99 Not applicable
  1 Retired
  2 Employed or self-employed
  3 Unemployed
  4 Permanently sick
  5 Homemaker
  97 Other
```

```
73. replace cjs=6 if cjs==97
(1,823 real changes made)
```

74. label define cjsLab 6 "Other", add

75. label values cjs cjsLab

76. tabulate cjs, nolab

Current job situation	Freq.	Percent	Cum.
1	73,119	66.44	66.44
2	20,471	18.60	85.05
3	1,877	1.71	86.75
4	2,700	2.45	89.21
5	10,055	9.14	98.34
6	1,823	1.66	100.00
Total	110,045	100.00	

77.

78. bysort id: egen Mundlak_OthC_age=mean(age)

79. bysort id: egen Mundlak_OthC_yedu=mean(yedu)

80. bysort id: egen Mundlak_OthC_bmi=mean(bmi)

81. bysort id: egen Mundlak_OthC_eurod=mean(eurod)

82.

83.

84.

85. *program for calculating Mundlak_Means

86. program define Mundlak_Means

1. *syntax varname

87. global variab changed schanged mst single sphus gali cjs thinc

2. foreach V of numlist 1(1)8 {

3. local va: word `V' of \$variab

4. tab `va', gen(_dum_)

5. levelsof `va', local(levels)

6. local i = 1

7. foreach l of local levels {

8. local lbl : label (`va') `l'

9. local clean = ustrword("`lbl'",1)

10. rename _dum_`i' `va'`clean'

11. local ++i

12. }

13. foreach v of varlist `va'_* {

14. if `v'==1 {

15. bysort id: egen Mundlak_C_`v' = mean(`v')

>

16. }

17. if `v'==2 {

18. bysort id: egen Mundlak_sC_`v' = mean(`v')

>

19. }

20. if `v'==3 {

21. bysort id: egen Mundlak_Mar_`v' = mean(`v')

>

22. }

23. if `v'==4 {

24. bysort id: egen Mundlak_Sin_`v' = mean(`v')

>

25. }

26. if `v'==5 {

27. bysort id: egen Mundlak_Sph_`v' = mean(`v')

>

28. }

29. if `v'==6 {

30. bysort id: egen Mundlak_Gal_`v' = mean(`v')

>

31. }

32. if `v'==7 | `v'==8 {

33. bysort id: egen Mundlak_Oth_`v' = mean(`v')

```

34.                                     }
35.                                     }
36.      }
37. end

```

88.

89. Mundlak_Means

changed	Freq.	Percent	Cum.
No longer officially together	2,005	1.82	1.82
All other situations	107,712	97.88	99.70
Newly officially together	328	0.30	100.00
Total	110,045	100.00	

0 1 2

schanged	Freq.	Percent	Cum.
Hooked up	429	0.39	0.39
Unchanged	106,828	97.08	97.47
Back single	2,788	2.53	100.00
Total	110,045	100.00	

0 1 2

mst	Freq.	Percent	Cum.
Officially with partner	76,018	69.08	69.08
Away from partner	10,875	9.88	78.96
Never married	5,967	5.42	84.38
Lost partner	17,185	15.62	100.00
Total	110,045	100.00	

0 1 2 3

Single	Freq.	Percent	Cum.
No	77,574	70.49	70.49
Yes	32,471	29.51	100.00
Total	110,045	100.00	

0 1

Self-perceived health - US scale	Freq.	Percent	Cum.
Excellent	7,401	6.73	6.73
Very good	21,370	19.42	26.14
Good	42,951	39.03	65.18
Fair	28,702	26.08	91.26
Poor	9,621	8.74	100.00
Total	110,045	100.00	

1 2 3 4 5

Limitation with activities	Freq.	Percent	Cum.
Not limited	60,290	54.79	54.79
Limited	49,755	45.21	100.00
Total	110,045	100.00	

0 1

Current job situation	Freq.	Percent	Cum.
Retired	73,119	66.44	66.44
Employed or self-employed	20,471	18.60	85.05
Unemployed	1,877	1.71	86.75
Permanently sick	2,700	2.45	89.21
Homemaker	10,055	9.14	98.34
Other	1,823	1.66	100.00
Total	110,045	100.00	

1 2 3 4 5 6

thinc	Freq.	Percent	Cum.
No income	1,712	1.56	1.56
Below country average income	66,361	60.30	61.86
Above country average income	41,972	38.14	100.00
Total	110,045	100.00	

0 1 2

```

90.
91. generate iG1=1.changed_No#eurod
92. generate iG2=1.changed_All#eurod
93. generate iG3=1.changed_Newly#eurod
94.
95. generate isG1=1.schanged_Hooked#eurod
96. generate isG2=1.schanged_Unchanged#eurod
97. generate isG3=1.schanged_Back#eurod
98.
99. foreach n of numlist 1(1)3 {
    2.      bysort id: egen Mundlak_Gi_`n'=mean(iG`n')
    3. }

100
101 foreach n of numlist 1(1)3 {
    2.      bysort id: egen Mundlak_sGi_`n'=mean(isG`n')
    3. }

102
103
104 ***descriptives
105
106 asdoc codebook phinact sphus thinc mst changed schanged cjs age yedu gender country
    > bmi eurod, compact save(report.doc) append

```

Variable	Obs	Unique	Mean	Min	Max	Label
phinact	110045	2	.119233	0	1	Physical inactivity
sphus	110045	5	3.106974	1	5	Self-perceived health - US sca
> le						
thinc	110045	3	1.36585	0	2	
mst	110045	4	.6757599	0	3	
changed	110045	3	.9847608	0	2	
schanged	110045	3	1.021437	0	2	
cjs	110045	6	1.74206	1	6	Current job situation
age	110045	47	70.09174	50	96	Age
yedu	110045	30	11.02087	0	30	Years of education
gender	110045	2	1.560752	1	2	Gender
country	110045	12	18.21905	11	29	Country identifier
bmi	110045	3634	26.8345	12.86854	98.61933	Body mass index
eurod	110045	13	2.371121	0	12	EURO depression scale

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```

107
108 graph bar (count), over(mst, label(angle(90) labsize(small))) over(Qyear, label(angl
> e(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(small))
> ylabel(0(2500)40000,angle(0) labsize(small) grid)

109 graph export descrip1.png, replace height(2400)
file descrip1.png saved as PNG format

110 graph bar (count), over(single, label(angle(90) labsize(small))) over(Qyear, label(a
> ngle(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(smal
> l)) ylabel(0(2500)42500,angle(0) labsize(small) grid)

111 graph export descrip2.png, replace height(2400)
file descrip2.png saved as PNG format

112 graph bar (count), over(phinact, label(angle(90) labsize(small))) over(Qyear, label(
> angle(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(sma
> l)) ylabel(0(2500)52500,angle(0) labsize(small) grid)

113 graph export descrip3.png, replace height(2400)
file descrip3.png saved as PNG format

114
115 asdoc tabulate phinact, save(report.doc) append

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Physical inactivity	Freq.	Percent	Cum.
No	96,924	88.08	88.08
Yes	13,121	11.92	100.00
Total	110,045	100.00	

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```

116
117 ***regressions
118
119 xtset id Qyear, yearly

Panel variable: id (unbalanced)
Time variable: Qyear, 2015 to 2022, but with gaps
Delta: 1 year

120
121 xtlogit phinact age yedu i.gender i.thinc i.mst ibl.changed i.gali i.cjs c.bmi c.eur
> od i.country

Fitting comparison model:

Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33239.196
Iteration 2: Log likelihood = -31474.668
Iteration 3: Log likelihood = -31437.723
Iteration 4: Log likelihood = -31437.592
Iteration 5: Log likelihood = -31437.592

Fitting full model:

tau = 0.0 Log likelihood = -31437.592
tau = 0.1 Log likelihood = -31263.473
tau = 0.2 Log likelihood = -31098.19
tau = 0.3 Log likelihood = -30953.791
tau = 0.4 Log likelihood = -30850.606
tau = 0.5 Log likelihood = -30824.183
tau = 0.6 Log likelihood = -30940.648

```

Iteration 0: Log likelihood = **-30824.183**
 Iteration 1: Log likelihood = **-30271.77**
 Iteration 2: Log likelihood = **-30234.943**
 Iteration 3: Log likelihood = **-30234.55**
 Iteration 4: Log likelihood = **-30234.55**

Random-effects logistic regression
 Group variable: **id**

Random effects u_i ~ Gaussian

Number of obs = **110,045**
 Number of groups = **48,341**

Obs per group:
 min = **1**
 avg = **2.3**
 max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-30234.55**

Wald chi2(29) = **7017.88**
 Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.0955753	.0023314	40.99	0.000	.0910058
> .1001449	yedu	-.0288965	.0040766	-7.09	0.000	-.0368865
> -.0209065	gender Female	.0184915	.035508	0.52	0.603	-.0511028
> .0880859	thinc Below country average income	-.4534341	.0961682	-4.72	0.000	-.6419204
> -.2649478	Above country average income	-.6560687	.0990685	-6.62	0.000	-.8502394
> -.4618979	mst Away from partner	.1015575	.0585961	1.73	0.083	-.0132888
> .2164038	Never married	.3300291	.0717943	4.60	0.000	.1893149
> .4707433	Lost partner	.090229	.0451345	2.00	0.046	.0017671
> .178691	changed No longer officially together	-.2940148	.0923621	-3.18	0.001	-.4750411
> -.1129885	Newly officially together	.3953265	.2439247	1.62	0.105	-.0827571
> .8734101	gali Limited	1.417282	.033636	42.14	0.000	1.351357
> 1.483208	cjs Employed or self-employed	.2222638	.0602958	3.69	0.000	.1040861
> .3404414	Unemployed	.5877893	.1180766	4.98	0.000	.3563634
> .8192153	Permanently sick	1.690957	.0762829	22.17	0.000	1.541445
> 1.840469	Homemaker	.1891099	.0535201	3.53	0.000	.0842125
> .2940072	Other	.6676439	.0942118	7.09	0.000	.4829922
> .8522956	bmi	.0477364	.0031852	14.99	0.000	.0414936
> .0539792						

> .2049693	eurod	.192974	.0061202	31.53	0.000	.1809786
> -.0589427	country Germany	-.2318761	.0882329	-2.63	0.009	-.4048095
> -.7665399	Sweden	-.9586265	.0980052	-9.78	0.000	-1.150713
> 1.042715	Spain	.8843885	.0807802	10.95	0.000	.7260621
> 1.989613	Italy	1.835259	.0787537	23.30	0.000	1.680904
> .5087977	France	.3393567	.0864511	3.93	0.000	.1699158
> -.2419429	Denmark	-.435573	.0987927	-4.41	0.000	-.6292031
> .3945341	Greece	.2305772	.083653	2.76	0.006	.0666203
> -.589939	Switzerland	-.8006815	.1075237	-7.45	0.000	-1.011424
> .5678298	Belgium	.4085186	.0812827	5.03	0.000	.2492075
> .4321668	Czechia	.2685154	.0834972	3.22	0.001	.1048639
> 1.017698	Poland	.8288242	.0963659	8.60	0.000	.6399504
> -11.67617	_cons	-12.17746	.2557661	-47.61	0.000	-12.67876
> 1.095025	/lnsig2u	1.021505	.0375108			.9479856
> 1.728947	sigma_u	1.666545	.0312567			1.606395
> .4760627	rho	.4577654	.0093108			.4395812

LR test of rho=0: chibar2(01) = 2406.08

Prob >= chibar2 = 0.000

122 estimates store RE

123 xtlogit phinact age yedu i.gender i.thinc i.mst ib1.changed i.gali i.cjs c.bmi c.eur
> od i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -5823.4647

Iteration 1: Log likelihood = -5415.1378

Iteration 2: Log likelihood = -5413.4208

Iteration 3: Log likelihood = -5413.4207

Conditional fixed-effects logistic regression
Group variable: **id**Number of obs = 17,262
Number of groups = 5,962

Obs per group:
 min = 2
 avg = 2.9
 max = 4

Log likelihood = -5413.4207

LR chi2(17) = 1770.89
 Prob > chi2 = 0.0000

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.1761131	.0059858	29.42	0.000	.1643811
> .1878452	yedu	-.0400949	.0193493	-2.07	0.038	-.0780189
> -.0021709						
	gender					
	Female	0	(omitted)			
	thinc					
Below country average income		.1874996	.1259732	1.49	0.137	-.0594034
> .4344026						
Above country average income		.1437623	.1334624	1.08	0.281	-.1178191
> .4053437						
	mst					
Away from partner		-.1226939	.3383987	-0.36	0.717	-.7859431
> .5405553						
Never married		-.2117363	.5574305	-0.38	0.704	-1.30428
> .8808074						
Lost partner		.0300787	.1612926	0.19	0.852	-.2860489
> .3462063						
	changed					
No longer officially together		-.0384458	.1540652	-0.25	0.803	-.340408
> .2635165						
Newly officially together		-.2438509	.3543099	-0.69	0.491	-.9382856
> .4505838						
	gali					
Limited		.6927275	.0493872	14.03	0.000	.5959303
> .7895247						
	cjs					
Employed or self-employed		.5220355	.1022354	5.11	0.000	.3216579
> .7224131						
Unemployed		.5511421	.174407	3.16	0.002	.2093107
> .8929735						
Permanently sick		.7985226	.1227411	6.51	0.000	.5579545
> 1.039091						
Homemaker		.1647436	.0975357	1.69	0.091	-.0264229
> .3559101						
Other		.5360843	.1401564	3.82	0.000	.2613828
> .8107858						
	bmi	-.0061261	.0074079	-0.83	0.408	-.0206452
> .008393						
eurod		.1054005	.0090707	11.62	0.000	.0876222
> .1231788						
	country					
Germany		0	(omitted)			
Sweden		0	(omitted)			
Spain		0	(omitted)			
Italy		0	(omitted)			
France		0	(omitted)			
Denmark		0	(omitted)			
Greece		0	(omitted)			
Switzerland		0	(omitted)			

Belgium	0	(omitted)
Czechia	0	(omitted)
Poland	0	(omitted)

124 estimates store FE

125 asdoc hausman FE RE, allegs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
age	.1761131	.0955753	.0805378	.0055131
yedu	-.0400949	-.0288965	-.0111984	.018915
thinc				
1	.1874996	-.4534341	.6409336	.0813691
2	.1437623	-.6560687	.7998309	.0894295
mst				
1	-.1226939	.1015575	-.2242513	.3332869
2	-.2117363	.3300291	-.5417654	.5527878
3	.0300787	.090229	-.0601503	.1548489
changed				
0	-.0384458	-.2940148	.2555691	.1233099
2	-.2438509	.3953265	-.6391774	.2569752
1.gali	.6927275	1.417282	-.7245549	.0361624
cjs				
2	.5220355	.2222638	.2997717	.082562
3	.5511421	.5877893	-.0366472	.1283577
4	.7985226	1.690957	-.8924346	.0961576
5	.1647436	.1891099	-.0243662	.0815403
6	.5360843	.6676439	-.1315596	.1037688
bmi	-.0061261	.0477364	-.0538625	.0066881
eurod	.1054005	.192974	-.0875735	.0066949

b = Consistent under H0 and Ha; obtained from **xtlogit**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(17) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= 1113.01

Prob > chi2 = 0.0000

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126

127 xtlogit phinact age yedu i.gender i.thinc i.mst i.gali i.cjs c.bmi ib1.changed##c.eu
> rod i.country, fe or baselevels

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.

note: **2.gender** omitted because of no within-group variance.

note: **12.country** omitted because of no within-group variance.

note: **13.country** omitted because of no within-group variance.

note: **15.country** omitted because of no within-group variance.

note: **16.country** omitted because of no within-group variance.

note: **17.country** omitted because of no within-group variance.

note: **18.country** omitted because of no within-group variance.

note: **19.country** omitted because of no within-group variance.

note: **20.country** omitted because of no within-group variance.

note: **23.country** omitted because of no within-group variance.

note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -5822.4756

Iteration 1: Log likelihood = -5413.7224

Iteration 2: Log likelihood = -5411.9886

Iteration 3: Log likelihood = -5411.9885

Conditional fixed-effects logistic regression
Group variable: **id**

Number of obs = **17,262**
Number of groups = **5,962**

Obs per group:
min = **2**
avg = **2.9**
max = **4**

Log likelihood = **-5411.9885**

LR chi2(19) = **1773.76**
Prob > chi2 = **0.0000**

> interval]		phinact	Odds ratio	Std. err.	z	P> z	[95% conf.
		age	1.192475	.0071401	29.40	0.000	1.178562
> 1.206552		yedu	.9604761	.0185742	-2.09	0.037	.9247526
> .9975797		gender					
		Male	1	(base)			
		Female	1	(omitted)			
		thinc					
		No income	1	(base)			
		Below country average income	1.200751	.1511905	1.45	0.146	.9381566
> 1.536847		Above country average income	1.150029	.1534278	1.05	0.295	.8854179
> 1.493721		mst					
		Officially with partner	1	(base)			
		Away from partner	.8476568	.2900945	-0.48	0.629	.4334235
> 1.657783		Never married	.7825463	.4390316	-0.44	0.662	.2605918
> 2.349953		Lost partner	1.03798	.1678054	0.23	0.818	.7560996
> 1.424947		gali					
		Not limited	1	(base)			
		Limited	1.998207	.0987121	14.01	0.000	1.813806
> 2.201355		cjs					
		Retired	1	(base)			
		Employed or self-employed	1.686989	.172557	5.11	0.000	1.380528
> 2.06148		Unemployed	1.733926	.3025838	3.15	0.002	1.231655
> 2.441024		Permanently sick	2.218185	.272318	6.49	0.000	1.743809
> 2.821606		Homemaker	1.177361	.1148165	1.67	0.094	.9725242
> 1.425342		Other	1.711756	.2398554	3.84	0.000	1.300677
> 2.252756		bmi	.9937933	.0073609	-0.84	0.401	.9794704
> 1.008326		changed					
		No longer officially together	1.305515	.3142663	1.11	0.268	.8144767
> 2.092593		All other situations	1	(base)			
		Newly officially together	.6434043	.3517823	-0.81	0.420	.2203374
> 1.878796		eurod	1.113732	.0102544	11.70	0.000	1.093814
> 1.134013							

changed#c.eurod						
No longer officially together		.9380614	.0366094	-1.64	0.101	.8689839
> 1.01263						
Newly officially together		1.055049	.1270821	0.44	0.656	.8331904
> 1.335983						
country						
Austria		1	(base)			
Germany		1	(omitted)			
Sweden		1	(omitted)			
Spain		1	(omitted)			
Italy		1	(omitted)			
France		1	(omitted)			
Denmark		1	(omitted)			
Greece		1	(omitted)			
Switzerland		1	(omitted)			
Belgium		1	(omitted)			
Czechia		1	(omitted)			
Poland		1	(omitted)			

128 outreg2 using results.xls, excel replace dec(3) alpha(0.01, 0.05, 0.10) symbol(***,
> **, *) ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout

129

130 xtlogit phinact age yedu i.gender i.thinc i.mst i.gali i.cjs c.bmi ib1.changed##c.eu
> rod i.country ///
> Mundlak_Gal_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Gi_* Mundlak_Mar_* //
> /
> , or baselevels vce(robust)
note: **Mundlak_Gal_gali_Limited** omitted because of collinearity.
note: **Mundlak_C_changed_Newly** omitted because of collinearity.
note: **Mundlak_Oth_cjs_Other** omitted because of collinearity.
note: **Mundlak_Oth_thinc_Above** omitted because of collinearity.
note: **Mundlak_Gi_3** omitted because of collinearity.
note: **Mundlak_Mar_mst_Lost** omitted because of collinearity.

Fitting comparison model:

Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -32908.193
Iteration 2: Log pseudolikelihood = -30977.077
Iteration 3: Log pseudolikelihood = -30929.154
Iteration 4: Log pseudolikelihood = -30928.954
Iteration 5: Log pseudolikelihood = -30928.954

Fitting full model:

tau = 0.0 Log pseudolikelihood = -30928.954
tau = 0.1 Log pseudolikelihood = -30752.591
tau = 0.2 Log pseudolikelihood = -30584.542
tau = 0.3 Log pseudolikelihood = -30436.359
tau = 0.4 Log pseudolikelihood = -30327.755
tau = 0.5 Log pseudolikelihood = -30293.604
tau = 0.6 Log pseudolikelihood = -30399.538

Iteration 0: Log pseudolikelihood = -30293.604
Iteration 1: Log pseudolikelihood = -29733.557
Iteration 2: Log pseudolikelihood = -29676.502
Iteration 3: Log pseudolikelihood = -29675.592
Iteration 4: Log pseudolikelihood = -29675.591

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: id

Number of obs = 110,045
Number of groups = 48,341

Random effects u_i ~ Gaussian

Obs per group:
 min = 1
 avg = 2.3
 max = 4

Integration method: **mvaghermite**

Integration pts. = 12

Log pseudolikelihood = **-29675.591**

Wald chi2(50) = 6285.28
 Prob > chi2 = 0.0000

(Std. err. adjusted for 48,341 clust

> ers in id)

		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]							
		age	1.190884	.007201	28.89	0.000	1.176853
>	1.205081	yedu	.9568555	.0161646	-2.61	0.009	.9256921
>	.9890679						
		gender					
		Male	1 (base)				
		Female	.994508	.0378987	-0.14	0.885	.9229342
>	1.071632						
		thinc					
		No income	1 (base)				
		Below country average income	1.129012	.1605331	0.85	0.393	.8544117
>	1.491867						
		Above country average income	1.095724	.162287	0.62	0.537	.8196532
>	1.464779						
		mst					
		Officially with partner	1 (base)				
		Away from partner	.924772	.3224931	-0.22	0.823	.4668717
>	1.831773						
		Never married	.9955141	.4634828	-0.01	0.992	.3997169
>	2.479376						
		Lost partner	1.09936	.1723295	0.60	0.546	.8085564
>	1.494753						
		gali					
		Not limited	1 (base)				
		Limited	1.996812	.0925232	14.92	0.000	1.82346
>	2.186643						
		cjs					
		Retired	1 (base)				
		Employed or self-employed	1.689087	.1745257	5.07	0.000	1.379435
>	2.068249						
		Unemployed	1.635531	.310341	2.59	0.010	1.127571
>	2.372323						
		Permanently sick	2.36124	.2958335	6.86	0.000	1.847121
>	3.018457						
		Homemaker	1.108253	.1189231	0.96	0.338	.8980477
>	1.367662						
		Other	1.696506	.2422325	3.70	0.000	1.282384
>	2.24436						
		bmi	.9865082	.0082107	-1.63	0.103	.9705461
>	1.002733						
		changed					
		No longer officially together	1.186845	.2668	0.76	0.446	.7639168
>	1.843921						
		All other situations	1 (base)				
		Newly officially together	.8369827	.4000308	-0.37	0.710	.3280121
>	2.135714						

	eurod	1.130377	.0109593	12.64	0.000	1.1091
>	1.152062					
	changed#c.eurod					
No longer officially together		.9555057	.0363734	-1.20	0.232	.8868097
>	1.029523					
	Newly officially together	1.032546	.1259643	0.26	0.793	.8129583
>	1.311448					
	country					
	Austria	1	(base)			
	Germany	.7130547	.0656178	-3.68	0.000	.5953773
>	.8539913					
	Sweden	.3877125	.0391072	-9.39	0.000	.3181647
>	.4724627					
	Spain	2.611382	.217744	11.51	0.000	2.21766
>	3.075005					
	Italy	6.613874	.544179	22.96	0.000	5.62886
>	7.77126					
	France	1.391673	.1249417	3.68	0.000	1.167127
>	1.65942					
	Denmark	.6268741	.0636759	-4.60	0.000	.51371
>	.764967					
	Greece	1.379181	.1216994	3.64	0.000	1.160142
>	1.639577					
	Switzerland	.5016099	.0563819	-6.14	0.000	.4024291
>	.6252343					
	Belgium	1.413895	.1189211	4.12	0.000	1.199012
>	1.667289					
	Czechia	1.24291	.1065378	2.54	0.011	1.050697
>	1.470285					
	Poland	2.163937	.2141602	7.80	0.000	1.782391
>	2.627158					
	Mundlak_Gal_gali_Not	.2535892	.0169918	-20.48	0.000	.22238
>	.2891784					
	Mundlak_Gal_gali_Limited	1	(omitted)			
	Mundlak_C_changed_No	.023174	.0260949	-3.34	0.001	.0025498
>	.2106157					
	Mundlak_C_changed_All	.1576523	.1641261	-1.77	0.076	.0204901
>	1.212986					
	Mundlak_C_changed Newly	1	(omitted)			
	Mundlak_Oth_cjs_Retired	.9187399	.1872665	-0.42	0.678	.6161585
>	1.369912					
	Mundlak_Oth_cjs_Employed	.6317673	.1460146	-1.99	0.047	.4016306
>	.9937738					
	Mundlak_Oth_cjs_Unemployed	.9823345	.3060849	-0.06	0.954	.5333769
>	1.809192					
	Mundlak_Oth_cjs_Permanently	2.666639	.6619143	3.95	0.000	1.639376
>	4.337603					
	Mundlak_Oth_cjs_Homemaker	.9650283	.2106805	-0.16	0.870	.6290856
>	1.480371					
	Mundlak_Oth_cjs_Other	1	(omitted)			
	Mundlak_Oth_thinc_No	5.431275	1.184453	7.76	0.000	3.542189
>	8.327831					
	Mundlak_Oth_thinc_Below	1.386007	.0987376	4.58	0.000	1.205388
>	1.593691					
	Mundlak_Oth_thinc_Above	1	(omitted)			
	Mundlak_OthC_age	.9132854	.0058062	-14.27	0.000	.901976
>	.9247366					
	Mundlak_OthC_yedu	1.024775	.0179003	1.40	0.161	.9902846
>	1.060466					
	Mundlak_OthC_bmi	1.07051	.0097293	7.50	0.000	1.05161
>	1.08975					
	Mundlak_OthC_eurod	.9085072	.2508532	-0.35	0.728	.5288082
>	1.560841					
	Mundlak_Gi_1	1.403464	.4028301	1.18	0.238	.7996218
>	2.463303					
	Mundlak_Gi_2	1.212639	.335168	0.70	0.485	.7054449
>	2.08449					
	Mundlak_Gi_3	1	(omitted)			

Mundlak_Mar_mst_Officially	1.041584	.1715723	0.25	0.805	.7541927
> 1.438488					
Mundlak_Mar_mst_Away	1.099025	.3851079	0.27	0.788	.5530176
> 2.184117					
Mundlak_Mar_mst_Never	1.273696	.605938	0.51	0.611	.501328
> 3.236008					
Mundlak_Mar_mst_Lost	1	(omitted)			
_cons	.0000628	.0000699	-8.69	0.000	7.08e-06
> .0005567					
/lnsig2u	1.096124	.0384112			1.020839
> 1.171409					
sigma_u	1.729897	.0332237			1.66599
> 1.796256					
rho	.4763368	.0095813			.4576001
> .4951404					

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

131 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)

results.xls

dir : seeout

132 asdoc, text(xtologitLim) save(report.doc) append

Click to Open File: report.doc

133 asdoc test Mundlak_Gal_gali_Not Mundlak_Gal_gali_Limited Mundlak_C_changed No Mundlak_C_changed All Mundlak_C_changed Newly Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs_Empl
> oyed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Homemake
> r Mundlak_Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below Mundlak_Oth_thi
> nc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak_OthC_eurod Mund
> lak_Gi_1 Mundlak_Gi_2 Mundlak_Gi_3 Mundlak_Mar_mst_Officially Mundlak_Mar_mst_Away M
> undlak_Mar_mst_Never Mundlak_Mar_mst_Lost, save(report.doc) append

```
( 1) [phinact]Mundlak_Gal_gali_Not = 0
( 2) [phinact]o.Mundlak_Gal_gali_Limited = 0
( 3) [phinact]Mundlak_C_changed_No = 0
( 4) [phinact]Mundlak_C_changed_All = 0
( 5) [phinact]o.Mundlak_C_changed_Newly = 0
( 6) [phinact]Mundlak_Oth_cjs_Retired = 0
( 7) [phinact]Mundlak_Oth_cjs_Employed = 0
( 8) [phinact]Mundlak_Oth_cjs_Unemployed = 0
( 9) [phinact]Mundlak_Oth_cjs_Permanently = 0
(10) [phinact]Mundlak_Oth_cjs_Homemaker = 0
(11) [phinact]o.Mundlak_Oth_cjs_Other = 0
(12) [phinact]Mundlak_Oth_thinc_No = 0
(13) [phinact]Mundlak_Oth_thinc_Below = 0
(14) [phinact]o.Mundlak_Oth_thinc_Above = 0
(15) [phinact]Mundlak_OthC_age = 0
(16) [phinact]Mundlak_OthC_yedu = 0
(17) [phinact]Mundlak_OthC_bmi = 0
(18) [phinact]Mundlak_OthC_eurod = 0
(19) [phinact]Mundlak_Gi_1 = 0
(20) [phinact]Mundlak_Gi_2 = 0
(21) [phinact]o.Mundlak_Gi_3 = 0
(22) [phinact]Mundlak_Mar_mst_Officially = 0
(23) [phinact]Mundlak_Mar_mst_Away = 0
(24) [phinact]Mundlak_Mar_mst_Never = 0
(25) [phinact]o.Mundlak_Mar_mst_Lost = 0
Constraint 2 dropped
Constraint 5 dropped
Constraint 11 dropped
Constraint 14 dropped
Constraint 21 dropped
Constraint 25 dropped
```

```
chi2( 19) = 966.58
Prob > chi2 = 0.0000
```

Click to Open File: [report.doc](#)

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```
135 logit phinact age yedu i.gender i.thinc i.mst i.gali i.cjs c.bmi ib1.changed##c.euro
> d i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)
```

```
Iteration 0: Log pseudolikelihood = -1.011e+08
Iteration 1: Log pseudolikelihood = -84670512
Iteration 2: Log pseudolikelihood = -81005214
Iteration 3: Log pseudolikelihood = -80947926
Iteration 4: Log pseudolikelihood = -80947802
Iteration 5: Log pseudolikelihood = -80947802
```

Logistic regression

```
Number of obs = 110,007
Wald chi2(34) = 4755.92
Prob > chi2 = 0.0000
Pseudo R2 = 0.1992
```

Log pseudolikelihood = -80947802

(Std. err. adjusted for 48,310 clust

> ers in id)

		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]							
		age	1.059557	.0033179	18.47	0.000	1.053074
>	1.06608	yedu	.9777868	.00535	-4.11	0.000	.967357
>	.9883292	gender					
		Male	1 (base)				
		Female	.9882358	.0445793	-0.26	0.793	.9046132
>	1.079588	thinc					
		No income	1 (base)				
		Below country average income	.5367557	.0798976	-4.18	0.000	.4009341
>	.7185887	Above country average income	.4376391	.0663829	-5.45	0.000	.3250891
>	.5891554	mst					
		Officially with partner	1 (base)				
		Away from partner	1.262653	.1087642	2.71	0.007	1.066503
>	1.494879	Never married	1.213222	.1070236	2.19	0.028	1.020591
>	1.44221	Lost partner	1.108644	.0593565	1.93	0.054	.9982028
>	1.231303	gali					
		Not limited	1 (base)				
		Limited	2.902283	.1255669	24.63	0.000	2.666322
>	3.159125	cjs					
		Retired	1 (base)				
		Employed or self-employed	1.016471	.084854	0.20	0.845	.8630531
>	1.197161	Unemployed	1.74188	.2782515	3.47	0.001	1.273637
>	2.38227	Permanently sick	3.573798	.3696004	12.32	0.000	2.918094
>	4.376842	Homemaker	1.154182	.0776852	2.13	0.033	1.011538
>	1.316942	Other	1.424528	.1594382	3.16	0.002	1.143936
>	1.773946						

	bmi	1.038227	.0046449	8.39	0.000	1.029163
>	1.047371					
	changed					
No longer officially together		1.048389	.2474198	0.20	0.841	.6601445
>	1.664969					
	All other situations	1	(base)			
Newly officially together		1.348295	.5580478	0.72	0.470	.5990735
>	3.03452					
	eurod	1.145244	.0089839	17.29	0.000	1.12777
>	1.162988					
	changed#c.eurod					
No longer officially together		.9693542	.0373145	-0.81	0.419	.8989099
>	1.045319					
Newly officially together		.9678382	.0899348	-0.35	0.725	.8066893
>	1.161179					
	country					
	Austria	1	(base)			
>	Germany	.9438142	.0801643	-0.68	0.496	.7990766
	Sweden	.4871687	.0501152	-6.99	0.000	.3982135
>	.5959952					
	Spain	2.194203	.2010543	8.58	0.000	1.833502
>	2.625863					
	Italy	4.282793	.3233891	19.26	0.000	3.693634
>	4.965928					
	France	1.318101	.1050929	3.46	0.001	1.12741
>	1.541045					
	Denmark	.7815697	.0722492	-2.67	0.008	.6520511
>	.9368148					
	Greece	1.297809	.101427	3.34	0.001	1.113493
>	1.512636					
	Switzerland	.5959958	.061844	-4.99	0.000	.4863149
>	.7304135					
	Belgium	1.315297	.1034196	3.49	0.000	1.127445
>	1.534449					
	Czechia	1.192624	.1242925	1.69	0.091	.9722848
>	1.462897					
	Poland	1.792868	.1576611	6.64	0.000	1.509022
>	2.130106					
	Qyear					
	2015	1	(base)			
>	.8714188	.7973703	.0361279	-5.00	0.000	.7296141
	2017					
>	1.057632	.9640212	.0455826	-0.77	0.438	.8786957
	2020					
>	1.079519	.9908455	.0433312	-0.21	0.833	.9094557
	2022					
>						
	_cons	.0004148	.000138	-23.40	0.000	.000216
>	.0007963					

Note: **_cons** estimates baseline odds.

```

136 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results.xls
  dir : seeout

```

```

137

```

```

138

```

```

139 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.changed c.
> eurod i.country

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33239.64
Iteration 2: Log likelihood = -31480.962
Iteration 3: Log likelihood = -31444.267
Iteration 4: Log likelihood = -31444.137
Iteration 5: Log likelihood = -31444.137

```

Fitting full model:

```

tau = 0.0 Log likelihood = -31444.137
tau = 0.1 Log likelihood = -31269.897
tau = 0.2 Log likelihood = -31104.505
tau = 0.3 Log likelihood = -30960.02
tau = 0.4 Log likelihood = -30856.786
tau = 0.5 Log likelihood = -30830.367
tau = 0.6 Log likelihood = -30946.906

```

```

Iteration 0: Log likelihood = -30830.367
Iteration 1: Log likelihood = -30277.666
Iteration 2: Log likelihood = -30240.897
Iteration 3: Log likelihood = -30240.505
Iteration 4: Log likelihood = -30240.505

```

Random-effects logistic regression
Group variable: **id**

Number of obs = **110,045**
Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-30240.505**

Wald chi2(27) = **7020.04**
Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.0944435	.0022743	41.53	0.000	.0899859
> .098901	yedu	-.0286127	.0040701	-7.03	0.000	-.0365899
> -.0206356	gender					
	Female	.0069056	.0353916	0.20	0.845	-.0624607
> .0762719	thinc					
	Below country average income	-.4543356	.0961547	-4.73	0.000	-.6427953
> -.2658758	Above country average income	-.6550089	.0992557	-6.60	0.000	-.8495465
> -.4604713	single					
	Yes	.1261672	.0368881	3.42	0.001	.0538679
> .1984665						

>	1.484227	gali Limited	1.418308	.033633	42.17	0.000	1.352388
>	.3401961	cjs Employed or self-employed	.2220563	.0602765	3.68	0.000	.1039166
>	.8242896	Unemployed	.5928962	.11806	5.02	0.000	.3615029
>	1.848584	Permanently sick	1.699129	.0762539	22.28	0.000	1.549674
>	.2876202	Homemaker	.1828976	.0534309	3.42	0.001	.0781749
>	.8443877	Other	.6599053	.0941254	7.01	0.000	.4754229
>	.0537099	bmi	.0474716	.0031828	14.91	0.000	.0412334
No longer officially together	> -.1382038	changed together	-.3149643	.0901855	-3.49	0.000	-.4917247
>	.8359426	Newly officially together	.3578835	.2439122	1.47	0.142	-.1201756
>	.2046442	eurod	.1926466	.0061213	31.47	0.000	.180649
>	-.062946	country Germany	-.2358335	.0882095	-2.67	0.008	-.408721
>	-.7658648	Sweden	-.9579178	.0979881	-9.78	0.000	-1.149971
>	1.040105	Spain	.8820364	.0806485	10.94	0.000	.7239682
>	1.984186	Italy	1.830233	.078549	23.30	0.000	1.67628
>	.5094096	France	.3400466	.0864113	3.94	0.000	.1706836
>	-.2409093	Denmark	-.4345574	.0988019	-4.40	0.000	-.6282055
>	.3870298	Greece	.2233083	.0835329	2.67	0.008	.0595868
>	-.5886102	Switzerland	-.7993548	.1075248	-7.43	0.000	-1.010099
>	.5625554	Belgium	.4032893	.0812597	4.96	0.000	.2440232
>	.4202294	Czechia	.2568385	.0833642	3.08	0.002	.0934477
>	1.003433	Poland	.8151587	.09606	8.49	0.000	.6268847
>	-11.58382	_cons	-12.07633	.2512854	-48.06	0.000	-12.56884
>	1.094728	/lnsig2u	1.021233	.0374984			.9477377
>	1.728691	sigma_u	1.666318	.0312421			1.606196
>	.4759887	rho	.4576978	.0093075			.4395201

LR test of rho=0: chibar2(01) = 2407.26

Prob >= chibar2 = 0.000

140 estimates store RE

141 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.changed c.
> eurod i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.

note: **2.gender** omitted because of no within-group variance.

note: **12.country** omitted because of no within-group variance.

note: **13.country** omitted because of no within-group variance.

note: **15.country** omitted because of no within-group variance.

note: **16.country** omitted because of no within-group variance.

note: **17.country** omitted because of no within-group variance.

note: **18.country** omitted because of no within-group variance.

note: **19.country** omitted because of no within-group variance.

note: **20.country** omitted because of no within-group variance.

note: **23.country** omitted because of no within-group variance.

note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-5826.2722**

Iteration 1: Log likelihood = **-5415.1119**

Iteration 2: Log likelihood = **-5413.4211**

Iteration 3: Log likelihood = **-5413.421**

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = **17,262**

Number of groups = **5,962**

Obs per group:

min = **2**

avg = **2.9**

max = **4**

Log likelihood = **-5413.421**

LR chi2(15) = **1770.89**

Prob > chi2 = **0.0000**

		phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]							
	age		.1771974	.0060019	29.52	0.000	.1654339
> .188961	yedu		-.039822	.0193161	-2.06	0.039	-.0776809
> -.0019631							
	gender						
	Female		0	(omitted)			
	thinc						
	Below country average income		.1864671	.1259642	1.48	0.139	-.0604182
> .4333524	Above country average income		.1387266	.1335789	1.04	0.299	-.1230831
> .4005364							
	single						
	Yes		-.0610691	.1106247	-0.55	0.581	-.2778896
> .1557513							
	gali						
	Limited		.6926735	.0493889	14.02	0.000	.595873
> .789474							
	cjs						
	Employed or self-employed		.5234409	.1021465	5.12	0.000	.3232375
> .7236443	Unemployed		.5549879	.174439	3.18	0.001	.2130938
> .896882	Permanently sick		.8016875	.1227456	6.53	0.000	.5611105
> 1.042264	Homemaker		.1631517	.097527	1.67	0.094	-.0279977

> .3543011						
> .8135211	Other	.5388196	.1401564	3.84	0.000	.2641182
> .0082688	bmi	-.0062586	.0074121	-0.84	0.398	-.0207859
No longer officially together	changed	.0165795	.1303156	0.13	0.899	-.2388343
> .2719933	Newly officially together	-.1837052	.3042401	-0.60	0.546	-.7800048
> .4125943						
> .1231945	eurod	.1054186	.0090695	11.62	0.000	.0876426
	country					
	Germany	0	(omitted)			
	Sweden	0	(omitted)			
	Spain	0	(omitted)			
	Italy	0	(omitted)			
	France	0	(omitted)			
	Denmark	0	(omitted)			
	Greece	0	(omitted)			
	Switzerland	0	(omitted)			
	Belgium	0	(omitted)			
	Czechia	0	(omitted)			
	Poland	0	(omitted)			

142 estimates store FE

143 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	$\sqrt{\text{diag}(V_b - V_B)}$ Std. err.
age	.1771974	.0944435	.082754	.0055543
yedu	-.039822	-.0286127	-.0112092	.0188824
thinc				
1	.1864671	-.4543356	.6408027	.0813711
2	.1387266	-.6550089	.7937356	.0893959
1.single	-.0610691	.1261672	-.1872363	.1042933
1.gali	.6926735	1.418308	-.7256342	.0361674
cjs				
2	.5234409	.2220563	.3013846	.082466
3	.5549879	.5928962	-.0379083	.1284165
4	.8016875	1.699129	-.8974414	.0961864
5	.1631517	.1828976	-.0197459	.0815883
6	.5388196	.6599053	-.1210857	.1038471
bmi	-.0062586	.0474716	-.0537302	.0066939
changed				
0	.0165795	-.3149643	.3315438	.0940676
2	-.1837052	.3578835	-.5415888	.1818485
eurod	.1054186	.1926466	-.0872281	.0066922

b = Consistent under H0 and Ha; obtained from **xtlogit**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

$\chi^2(15) = (b-B)'[(V_b - V_B)^{-1}](b-B)$
 = 1117.62
 Prob > χ^2 = 0.0000
 Click to Open File: [report.doc](#)

144

145

146 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.schanged##
 > c.eurod i.country, fe or baselevels

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
 all negative outcomes.

note: **2.gender** omitted because of no within-group variance.

note: **12.country** omitted because of no within-group variance.

note: **13.country** omitted because of no within-group variance.

note: **15.country** omitted because of no within-group variance.

note: **16.country** omitted because of no within-group variance.

note: **17.country** omitted because of no within-group variance.

note: **18.country** omitted because of no within-group variance.

note: **19.country** omitted because of no within-group variance.

note: **20.country** omitted because of no within-group variance.

note: **23.country** omitted because of no within-group variance.

note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-5822.8804**

Iteration 1: Log likelihood = **-5411.5765**

Iteration 2: Log likelihood = **-5409.7982**

Iteration 3: Log likelihood = **-5409.7981**

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = **17,262**

Number of groups = **5,962**

Obs per group:

min = **2**

avg = **2.9**

max = **4**

LR chi2(17) = **1778.14**

Prob > chi2 = **0.0000**

Log likelihood = **-5409.7981**

		phinact	Odds ratio	Std. err.	z	P> z	[95% conf.
> interval]							
		age	1.193982	.0072111	29.36	0.000	1.179932
> 1.208199		yedu	.9611865	.0185572	-2.05	0.040	.9254946
> .9982549		gender					
		Male	1	(base)			
		Female	1	(omitted)			
		thinc					
		No income	1	(base)			
Below country average income			1.199121	.1509844	1.44	0.149	.9368845
> 1.534758		Above country average income	1.145213	.1529077	1.02	0.310	.8815247
> 1.487777		single					
		No	1	(base)			
> 1.354112		Yes	1.027117	.1448418	0.19	0.850	.7790858
		gali					
		Not limited	1	(base)			
> 2.199543		Limited	1.996492	.0986631	13.99	0.000	1.812185
		cjs					
		Retired	1	(base)			
Employed or self-employed			1.699259	.1735673	5.19	0.000	1.390962
> 2.075888							

	Unemployed	1.744534	.3045186	3.19	0.001	1.239074
> 2.456188						
	Permanently sick	2.217455	.2724094	6.48	0.000	1.742956
> 2.821129						
	Homemaker	1.177278	.1148587	1.67	0.094	.9723742
> 1.425361						
	Other	1.719782	.2410161	3.87	0.000	1.306722
> 2.263412						
	bmi	.9939817	.0073702	-0.81	0.416	.9796408
> 1.008533						
	schanged					
	Hooked up	.4582858	.2150017	-1.66	0.096	.1827262
> 1.149402						
	Unchanged	1	(base)			
	Back single	1.130955	.2278444	0.61	0.541	.7620101
> 1.678534						
	eurod	1.114476	.0103228	11.70	0.000	1.094426
> 1.134893						
	schanged#c.eurod					
	Hooked up	1.197643	.1373797	1.57	0.116	.9565052
> 1.499572						
	Back single	.9449816	.0309743	-1.73	0.084	.8861821
> 1.007683						
	country					
	Austria	1	(base)			
	Germany	1	(omitted)			
	Sweden	1	(omitted)			
	Spain	1	(omitted)			
	Italy	1	(omitted)			
	France	1	(omitted)			
	Denmark	1	(omitted)			
	Greece	1	(omitted)			
	Switzerland	1	(omitted)			
	Belgium	1	(omitted)			
	Czechia	1	(omitted)			
	Poland	1	(omitted)			

```

147 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout

```

```

148
149 xtlogit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ibl.schanged##
> c.eurod i.country ///
> Mundlak_Gal_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_sGi_* Mundlak_Sin_*
> ///
> , or baselevels vce(robust)
note: Mundlak_Gal_gali_Limited omitted because of collinearity.
note: Mundlak_sC_schanged_Back omitted because of collinearity.
note: Mundlak_Oth_cjs_Other omitted because of collinearity.
note: Mundlak_Oth_thinc_Above omitted because of collinearity.
note: Mundlak_sGi_3 omitted because of collinearity.
note: Mundlak_Sin_single_Yes omitted because of collinearity.

```

Fitting comparison model:

```

Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -32915.544
Iteration 2: Log pseudolikelihood = -30990.562
Iteration 3: Log pseudolikelihood = -30942.807
Iteration 4: Log pseudolikelihood = -30942.608
Iteration 5: Log pseudolikelihood = -30942.608

```

Fitting full model:

```
tau = 0.0    Log pseudolikelihood = -30942.608
tau = 0.1    Log pseudolikelihood = -30766.122
tau = 0.2    Log pseudolikelihood = -30597.883
tau = 0.3    Log pseudolikelihood = -30449.443
tau = 0.4    Log pseudolikelihood = -30340.513
tau = 0.5    Log pseudolikelihood = -30305.96
tau = 0.6    Log pseudolikelihood = -30411.387
```

```
Iteration 0: Log pseudolikelihood = -30305.961
Iteration 1: Log pseudolikelihood = -29745.478
Iteration 2: Log pseudolikelihood = -29687.607
Iteration 3: Log pseudolikelihood = -29686.678
Iteration 4: Log pseudolikelihood = -29686.677
```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Number of obs = 110,045
Number of groups = 48,341

Random effects u_i ~ Gaussian

Obs per group:
min = 1
avg = 2.3
max = 4

Integration method: **mvaghermite**

Integration pts. = 12

Log pseudolikelihood = **-29686.677**

Wald chi2(46) = 6277.20
Prob > chi2 = 0.0000

(Std. err. adjusted for 48,341 clusters)

> rs in id)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]						
	age	1.192276	.0072762	28.82	0.000	1.1781
> 1.206623	yedu	.9567055	.0161961	-2.61	0.009	.9254826
> .9889818	gender					
	Male	1 (base)				
	Female	.9852918	.0376216	-0.39	0.698	.9142465
> 1.061858	thinc					
	No income	1 (base)				
	Below country average income	1.124765	.160014	0.83	0.409	.8510713
> 1.486474	Above country average income	1.090266	.1617837	0.58	0.560	.8151229
> 1.458282	single					
	No	1 (base)				
	Yes	1.059423	.1473351	0.42	0.678	.8066626
> 1.391384	gali					
	Not limited	1 (base)				
	Limited	1.997384	.0925604	14.93	0.000	1.823964
> 2.187293	cjs					
	Retired	1 (base)				
	Employed or self-employed	1.697181	.1755446	5.11	0.000	1.385753
> 2.078598	Unemployed	1.643797	.3125556	2.61	0.009	1.132397

>	2.38615						
>	3.026963	Permanently sick	2.36741	.2968536	6.87	0.000	1.851569
>	1.369455	Homemaker	1.109721	.1190732	0.97	0.332	.8992485
>	2.250535	Other	1.70085	.2430178	3.72	0.000	1.285423
>	1.002837	bmi	.9866044	.0082145	-1.62	0.105	.9706349
>	1.153011	schanged Hooked up	.4039607	.2161671	-1.69	0.090	.1415288
>	1.620576	Unchanged Back single	1 (base) 1.093877	.2193675	0.45	0.655	.7383584
>	1.153424	eurod	1.131628	.0110147	12.70	0.000	1.110245
>	1.56395	schanged#c.eurod Hooked up	1.256824	.1401944	2.05	0.040	1.01001
>	1.019812	Back single	.9532018	.0328504	-1.39	0.164	.8909426
>	.856164	country Austria Germany	1 (base) .7148439	.0657952	-3.65	0.000	.5968503
>	.4721821	Sweden	.3874259	.039107	-9.39	0.000	.3178833
>	3.082666	Spain	2.61847	.2180377	11.56	0.000	2.224175
>	7.792686	Italy	6.634219	.5447764	23.04	0.000	5.64797
>	1.664248	France	1.395815	.125266	3.72	0.000	1.170678
>	.7667265	Denmark	.6280886	.0639152	-4.57	0.000	.514519
>	1.633738	Greece	1.374548	.1211488	3.61	0.000	1.156478
>	.6259042	Switzerland	.5021654	.0564351	-6.13	0.000	.4028893
>	1.674062	Belgium	1.419492	.1194666	4.16	0.000	1.203635
>	1.460673	Czechia	1.23519	.1056687	2.47	0.014	1.044515
>	2.623332	Poland	2.16212	.2132992	7.82	0.000	1.781995
>	.288651	Mundlak_Gal_gali_Not	.2531119	.0169674	-20.50	0.000	.2219484
>	39.51996	Mundlak_Gal_gali_Limited Mundlak_sC_schanged_Hooked	1 (omitted) 5.542075	5.554723	1.71	0.088	.777192
>	5.419336	Mundlak_sC_schanged_Unchanged	2.593802	.9751401	2.54	0.011	1.241445
>	1.365591	Mundlak_sC_schanged_Back Mundlak_Oth_cjs_ReTired	1 (omitted) .9159816	.1866328	-0.43	0.667	.6144025
>	.9885426	Mundlak_Oth_cjs_Employed	.6286265	.1451944	-2.01	0.044	.3997514
>	1.817235	Mundlak_Oth_cjs_Unemployed	.9872047	.3073456	-0.04	0.967	.5362945
>	4.390044	Mundlak_Oth_cjs_Permanently	2.69805	.6701334	4.00	0.000	1.658178
>	1.473654	Mundlak_Oth_cjs_Homemaker	.9604082	.2097953	-0.18	0.853	.6259161
>		Mundlak_Oth_cjs_Other Mundlak_Oth_thInc No	1 (omitted) 5.356837	1.170752	7.68	0.000	3.490405

> 8.221309	Mundlak_Oth_thinc_Below	1.377161	.0996741	4.42	0.000	1.195027
> 1.587054	Mundlak_Oth_thinc_Above	1 (omitted)				
	Mundlak_OthC_age	.9117775	.0058295	-14.45	0.000	.9004232
> .923275	Mundlak_OthC_yedu	1.024981	.0179354	1.41	0.159	.990424
> 1.060743	Mundlak_OthC_bmi	1.070222	.0097335	7.46	0.000	1.051314
> 1.089471	Mundlak_OthC_eurod	1.173928	.0826663	2.28	0.023	1.022589
> 1.347665	Mundlak_sGi_1	.5933988	.1474502	-2.10	0.036	.3646166
> .9657324	Mundlak_sGi_2	.9383699	.0670739	-0.89	0.374	.8157008
> 1.079487	Mundlak_sGi_3	1 (omitted)				
	Mundlak_Sin_single_No	1.007241	.1467563	0.05	0.961	.7570285
> 1.340153	Mundlak_Sin_single_Yes	1 (omitted)				
	_cons	4.17e-06	2.25e-06	-22.99	0.000	1.45e-06
> .000012						
	/lnsig2u	1.098203	.0383644			1.02301
> 1.173396						
	sigma_u	1.731696	.0332177			1.667799
> 1.798041						
	rho	.4768554	.0095705			.4581388
> .4956371						

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

```
150 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout
```

```
151 asdoc, text(xtologitLim) save(report.doc) append
Click to Open File: report.doc
```

```
152 asdoc test Mundlak_Gal_gali_Not Mundlak_Gal_gali_Limited Mundlak_sC_schanged_Hooked
> Mundlak_sC_schanged_Unchanged Mundlak_sC_schanged_Back Mundlak_Oth_cjs_Retired Mundlak_O
> ak_Oth_cjs_Employed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_O
> th_cjs_Homemaker Mundlak_Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below
> Mundlak_Oth_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak
> OthC_eurod Mundlak_sGi_1 Mundlak_sGi_2 Mundlak_sGi_3 Mundlak_Sin_single_No Mundlak_S
> in_single_Yes, save(report.doc) append
```

```
( 1) [phinact]Mundlak_Gal_gali_Not = 0
( 2) [phinact]o.Mundlak_Gal_gali_Limited = 0
( 3) [phinact]Mundlak_sC_schanged_Hooked = 0
( 4) [phinact]Mundlak_sC_schanged_Unchanged = 0
( 5) [phinact]o.Mundlak_sC_schanged_Back = 0
( 6) [phinact]Mundlak_Oth_cjs_Retired = 0
( 7) [phinact]Mundlak_Oth_cjs_Employed = 0
( 8) [phinact]Mundlak_Oth_cjs_Unemployed = 0
( 9) [phinact]Mundlak_Oth_cjs_Permanently = 0
(10) [phinact]Mundlak_Oth_cjs_Homemaker = 0
(11) [phinact]o.Mundlak_Oth_cjs_Other = 0
(12) [phinact]Mundlak_Oth_thinc_No = 0
(13) [phinact]Mundlak_Oth_thinc_Below = 0
(14) [phinact]o.Mundlak_Oth_thinc_Above = 0
(15) [phinact]Mundlak_OthC_age = 0
(16) [phinact]Mundlak_OthC_yedu = 0
(17) [phinact]Mundlak_OthC_bmi = 0
(18) [phinact]Mundlak_OthC_eurod = 0
(19) [phinact]Mundlak_sGi_1 = 0
```

```

(20) [phinact]Mundlak_sGi_2 = 0
(21) [phinact]o.Mundlak_sGi_3 = 0
(22) [phinact]Mundlak_Sin_single_No = 0
(23) [phinact]o.Mundlak_Sin_single_Yes = 0
Constraint 2 dropped
Constraint 5 dropped
Constraint 11 dropped
Constraint 14 dropped
Constraint 21 dropped
Constraint 23 dropped

```

```

chi2( 17) = 950.32
Prob > chi2 = 0.0000

```

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153

```

154 logit phinact age yedu i.gender i.thinc i.single i.gali i.cjs c.bmi ib1.schanged#c.
> eurod i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

```

```

Iteration 0: Log pseudolikelihood = -1.011e+08
Iteration 1: Log pseudolikelihood = -84642684
Iteration 2: Log pseudolikelihood = -80978284
Iteration 3: Log pseudolikelihood = -80920915
Iteration 4: Log pseudolikelihood = -80920786
Iteration 5: Log pseudolikelihood = -80920786

```

Logistic regression

```

Number of obs = 110,007
Wald chi2(32) = 4747.55
Prob > chi2 = 0.0000
Pseudo R2 = 0.1995

```

Log pseudolikelihood = -80920786

(Std. err. adjusted for 48,310 clusters)

> rs in id)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]						
	age	1.057707	.0032545	18.23	0.000	1.051347
> 1.064105						
	yedu	.9779381	.0053284	-4.09	0.000	.9675503
> .9884375						
	gender					
	Male	1 (base)				
	Female	.9714813	.0439564	-0.64	0.523	.8890379
> 1.06157						
	thinc					
	No income	1 (base)				
	Below country average income	.5355138	.0793087	-4.22	0.000	.4005983
> .7158669						
	Above country average income	.4421286	.0665224	-5.42	0.000	.3292134
> .5937721						
	single					
	No	1 (base)				
	Yes	1.196428	.0592897	3.62	0.000	1.085687
> 1.318464						
	gali					
	Not limited	1 (base)				
	Limited	2.911047	.1258241	24.72	0.000	2.674593
> 3.168405						
	cjs					
	Retired	1 (base)				
	Employed or self-employed	1.015249	.0850707	0.18	0.857	.8614849
> 1.196457						
	Unemployed	1.750519	.277252	3.54	0.000	1.283368

>	2.387715	Permanently sick	3.583356	.3684595	12.41	0.000	2.929308
>	4.383439						
>	1.309347						
>	1.7384						
>	1.047163	bmi	1.038067	.0046206	8.39	0.000	1.029051
>	1.556769	schanged Hooked up	.6028327	.2918035	-1.05	0.296	.2334369
>	1.43184	Unchanged Back single	1 .9803691	(base) .1894679	-0.10	0.918	.6712509
>	1.163892	eurod	1.146075	.0090208	17.32	0.000	1.12853
>	1.188984	schanged#c.eurod Hooked up	.9667905	.102044	-0.32	0.749	.7861196
>	1.024871	Back single	.9589651	.0325211	-1.24	0.217	.8972971
>	1.112466	country Austria Germany	1 .941933	(base) .0799698	-0.70	0.481	.7975415
>	.5976023	Sweden	.4882099	.0503613	-6.95	0.000	.3988421
>	2.592304	Spain	2.167038	.1981174	8.46	0.000	1.811536
>	4.888928	Italy	4.222204	.3158434	19.25	0.000	3.646404
>	1.534022	France	1.312142	.104593	3.41	0.001	1.122355
>	.9397223	Denmark	.7838422	.0725374	-2.63	0.008	.6538194
>	1.491553	Greece	1.280307	.0997599	3.17	0.002	1.098979
>	.7328927	Switzerland	.5980315	.0620487	-4.96	0.000	.4879864
>	1.537887	Belgium	1.317725	.1038753	3.50	0.000	1.129081
>	1.451254	Czechia	1.183223	.1232662	1.61	0.106	.9646948
>	2.078159	Poland	1.751687	.1527419	6.43	0.000	1.476503
>	.8809112	Qyear 2015 2017	1 .8061496	(base) .0364779	-4.76	0.000	.7377329
>	1.072872	2020	.9776645	.0463539	-0.48	0.634	.8909061
>	1.09892	2022	1.008398	.0442286	0.19	0.849	.9253334
>	.000892	_cons	.0004715	.0001534	-23.55	0.000	.0002492

Note: **_cons** estimates baseline odds.

```

155 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results.xls
  dir : seeout

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156
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160
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```

161 xtlogit phinact age yedu i.gender i.thinc i.mst ib1.changed i.sphus i.cjs c.bmi c.eu
> rod i.country

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33341.273
Iteration 2: Log likelihood = -30402.042
Iteration 3: Log likelihood = -30159.285
Iteration 4: Log likelihood = -30158.011
Iteration 5: Log likelihood = -30158.011

```

Fitting full model:

```

tau = 0.0    Log likelihood = -30158.011
tau = 0.1    Log likelihood = -29999.182
tau = 0.2    Log likelihood = -29850.649
tau = 0.3    Log likelihood = -29724.821
tau = 0.4    Log likelihood = -29642.732
tau = 0.5    Log likelihood = -29641.353
tau = 0.6    Log likelihood = -29789.676

```

```

Iteration 0: Log likelihood = -29641.352
Iteration 1: Log likelihood = -29115.796
Iteration 2: Log likelihood = -29112.761
Iteration 3: Log likelihood = -29112.755
Iteration 4: Log likelihood = -29112.755

```

Random-effects logistic regression
Group variable: **id**

Number of obs = **110,045**
Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-29112.755**

Wald chi2(32) = **7815.66**
Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.0925454	.0022957	40.31	0.000	.0880459
> .0970448	yedu	-.0181854	.0040488	-4.49	0.000	-.0261208
> -.0102499	gender					
	Female	.1283035	.0352808	3.64	0.000	.0591545
> .1974526	thinc					
	Below country average income	-.570086	.0950744	-6.00	0.000	-.7564284
> -.3837435	Above country average income	-.7219872	.0978989	-7.37	0.000	-.9138655
> -.5301088						

		mst					
>	.1826511	Away from partner	.0683493	.0583183	1.17	0.241	-.0459525
>	.4525463	Never married	.3132433	.0710743	4.41	0.000	.1739403
>	.1978105	Lost partner	.1101811	.0447097	2.46	0.014	.0225517
		changed					
No longer officially together							
>	-.0609114		-.2435271	.093173	-2.61	0.009	-.4261427
>	1.035928	Newly officially together	.5536561	.2460617	2.25	0.024	.0713839
		sphus					
>	.1552382	Very good	-.0376238	.0984008	-0.38	0.702	-.2304859
>	.5275672	Good	.3490693	.0910721	3.83	0.000	.1705713
>	1.506107	Fair	1.326277	.0917519	14.46	0.000	1.146446
>	3.245931	Poor	3.054535	.0976529	31.28	0.000	2.863139
		cjs					
>	.4167737	Employed or self-employed	.2988579	.0601622	4.97	0.000	.1809421
>	.8372417	Unemployed	.6068024	.1175732	5.16	0.000	.3763631
>	1.587017	Permanently sick	1.43696	.076561	18.77	0.000	1.286904
>	.3001407	Homemaker	.1960264	.0531205	3.69	0.000	.0919121
>	.8041234	Other	.6191805	.0943603	6.56	0.000	.4342376
		bmi					
>	.0550295		.0488215	.0031674	15.41	0.000	.0426135
>	.1284691	eurod	.1161214	.0063	18.43	0.000	.1037737
		country					
>	-.1637156	Germany	-.3371138	.0884701	-3.81	0.000	-.5105121
>	-.7134061	Sweden	-.9056884	.098105	-9.23	0.000	-1.097971
>	.7528846	Spain	.5956352	.0802307	7.42	0.000	.4383859
>	1.817849	Italy	1.665115	.0779266	21.37	0.000	1.512382
>	.4058242	France	.2368863	.0861944	2.75	0.006	.0679485
>	-.2001903	Denmark	-.3938354	.0988003	-3.99	0.000	-.5874804
>	.1106427	Greece	-.0519691	.0829668	-0.63	0.531	-.214581
>	-.4657538	Switzerland	-.675283	.1069046	-6.32	0.000	-.8848122
>	.7242679	Belgium	.565509	.0810009	6.98	0.000	.4067501
>	.3323202	Czechia	.168629	.0835175	2.02	0.043	.0049378
>	.6687777	Poland	.4804049	.0961103	5.00	0.000	.2920322
>	-11.37568	_cons	-11.89399	.2644476	-44.98	0.000	-12.4123
>	.994412	/lnsig2u	.9185522	.0387047			.8426924

	sigma_u	1.582928	.0306334	1.524012
> 1.644121	rho	.4323437	.009499	.4138299
> .4510484				

LR test of rho=0: chibar2(01) = 2090.51

Prob >= chibar2 = 0.000

162 estimates store RE

163 xtlogit phinact age yedu i.gender i.thinc i.mst ibl.changed i.sphus i.cjs c.bmi c.eu

> rod i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or all negative outcomes.

note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -5682.4207

Iteration 1: Log likelihood = -5285.1486

Iteration 2: Log likelihood = -5283.173

Iteration 3: Log likelihood = -5283.1724

Iteration 4: Log likelihood = -5283.1724

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = 17,262

Number of groups = 5,962

Obs per group:

min = 2

avg = 2.9

max = 4

Log likelihood = -5283.1724

LR chi2(20) = 2031.39

Prob > chi2 = 0.0000

	phinact	Coefficient	Std. err.	z	P> z	[95% conf. interval]
> interval]						
	age	.1723345	.0060571	28.45	0.000	.1604628
> .1842062	yedu	-.0419938	.0195573	-2.15	0.032	-.0803255
> -.0036621						
	gender					
	Female	0	(omitted)			
	thinc					
	Below country average income	.1548825	.1266892	1.22	0.222	-.0934238
> .4031889	Above country average income	.108924	.1343678	0.81	0.418	-.1544321
> .3722801						
	mst					
	Away from partner	-.0502066	.338236	-0.15	0.882	-.7131371
> .6127239	Never married	-.1146535	.5812424	-0.20	0.844	-1.253868
> 1.024561	Lost partner	.0473318	.1626894	0.29	0.771	-.2715335

>	.3661971					
No longer officially together	changed					
>	.2500215					
Newly officially together						
>	.6181283					
	sphus					
>	.0752543	Very good				
	Good					
>	.2390798					
	Fair					
>	.7153047					
	Poor					
>	1.669639					
	cjs					
>	.708456	Employed or self-employed				
	Unemployed					
>	.8996334					
	Permanently sick					
>	1.008091					
	Homemaker					
>	.3272549					
	Other					
>	.7485944					
	bmi					
>	.011924					
	eurod					
>	.0940736					
	country					
	Germany	0	(omitted)			
	Sweden	0	(omitted)			
	Spain	0	(omitted)			
	Italy	0	(omitted)			
	France	0	(omitted)			
	Denmark	0	(omitted)			
	Greece	0	(omitted)			
	Switzerland	0	(omitted)			
	Belgium	0	(omitted)			
	Czechia	0	(omitted)			
	Poland	0	(omitted)			

164 estimates store FE

165 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
age	.1723345	.0925454	.0797891	.0056052
yedu	-.0419938	-.0181854	-.0238084	.0191337
thinc				
1	.1548825	-.570086	.7249685	.0837318
2	.108924	-.7219872	.8309112	.0920354
mst				
1	-.0502066	.0683493	-.1185559	.3331705
2	-.1146535	.3132433	-.4278968	.5768805
3	.0473318	.1101811	-.0628493	.1564253
changed				
0	-.0562219	-.2435271	.1873051	.1254301
2	-.0941338	.5536561	-.6477899	.2674272
sphus				
2	-.2058793	-.0376238	-.1682554	.1043636

3	-.0329768	.3490693	-.3820461	.1047533
4	.4373624	1.326277	-.8889143	.1081279
5	1.375587	3.054535	-1.678948	.113898
cjs				
2	.5062779	.2988579	.20742	.0837929
3	.5542604	.6068024	-.052542	.1312551
4	.7613586	1.43696	-.6756018	.0999285
5	.1339531	.1960264	-.0620733	.0830972
6	.4709706	.6191805	-.1482099	.1056414
bmi	-.0028905	.0488215	-.051712	.0068629
eurod	.0756036	.1161214	-.0405177	.0070082

b = Consistent under H0 and Ha; obtained from **xtlogit**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(20) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= **1180.10**
Prob > chi2 = **0.0000**
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```
167 xtlogit phinact age yedu i.gender i.thinc i.mst i.sphus i.cjs c.bmi ib1.changed##c.e
> urod i.country, fe or baselevels
note: multiple positive outcomes within groups encountered.
note: 42,379 groups (92,783 obs) omitted because of all positive or
      all negative outcomes.
note: 2.gender omitted because of no within-group variance.
note: 12.country omitted because of no within-group variance.
note: 13.country omitted because of no within-group variance.
note: 15.country omitted because of no within-group variance.
note: 16.country omitted because of no within-group variance.
note: 17.country omitted because of no within-group variance.
note: 18.country omitted because of no within-group variance.
note: 19.country omitted because of no within-group variance.
note: 20.country omitted because of no within-group variance.
note: 23.country omitted because of no within-group variance.
note: 28.country omitted because of no within-group variance.
note: 29.country omitted because of no within-group variance.
```

Iteration 0: Log likelihood = **-5681.7469**
Iteration 1: Log likelihood = **-5284.056**
Iteration 2: Log likelihood = **-5282.055**
Iteration 3: Log likelihood = **-5282.0544**
Iteration 4: Log likelihood = **-5282.0544**

Conditional fixed-effects logistic regression
Group variable: **id**

Number of obs = **17,262**
Number of groups = **5,962**

Obs per group:
min = **2**
avg = **2.9**
max = **4**

Log likelihood = **-5282.0544**
LR chi2(22) = **2033.63**
Prob > chi2 = **0.0000**

	phinact	Odds ratio	Std. err.	z	P> z	[95% conf.
> interval]						
	age	1.187998	.0071977	28.43	0.000	1.173974
> 1.202189	yedu	.9585801	.0187356	-2.16	0.030	.9225534
> .9960137						
	gender					
	Male	1	(base)			
	Female	1	(omitted)			

	thinc					
	No income	1	(base)			
	Below country average income	1.162598	.1472864	1.19	0.234	.9069709
>	1.490274					
	Above country average income	1.110737	.1492458	0.78	0.434	.8535686
>	1.445387					
	mst					
	Officially with partner	1	(base)			
	Away from partner	.9038505	.3101604	-0.29	0.768	.4613206
>	1.770885					
	Never married	.8497704	.4996188	-0.28	0.782	.2684364
>	2.690059					
	Lost partner	1.052871	.1716716	0.32	0.752	.7648675
>	1.449321					
	sphus					
	Excellent	1	(base)			
	Very good	.816106	.11707	-1.42	0.157	.616087
>	1.081063					
	Good	.9710693	.1348225	-0.21	0.833	.7397247
>	1.274766					
	Fair	1.553807	.2204464	3.11	0.002	1.17661
>	2.051924					
	Poor	3.966757	.5953088	9.18	0.000	2.955915
>	5.323278					
	cjs					
	Retired	1	(base)			
	Employed or self-employed	1.659968	.1713239	4.91	0.000	1.355963
>	2.03213					
	Unemployed	1.739401	.3065802	3.14	0.002	1.231318
>	2.457136					
	Permanently sick	2.138068	.2692534	6.03	0.000	1.670426
>	2.736629					
	Homemaker	1.142063	.1126234	1.35	0.178	.9413465
>	1.385576					
	Other	1.603559	.2271449	3.33	0.001	1.21482
>	2.116694					
	bmi	.9970378	.0075356	-0.39	0.695	.9823772
>	1.011917					
	changed					
	No longer officially together	1.224085	.3012522	0.82	0.411	.7556622
>	1.982877					
	All other situations	1	(base)			
	Newly officially together	.6756415	.3806081	-0.70	0.486	.2239805
>	2.038086					
	eurod	1.080515	.0103358	8.10	0.000	1.060446
>	1.100964					
	changed#c.eurod					
	No longer officially together	.947853	.0378779	-1.34	0.180	.8764467
>	1.025077					
	Newly officially together	1.087142	.1354772	0.67	0.503	.851553
>	1.38791					
	country					
	Austria	1	(base)			
	Germany	1	(omitted)			
	Sweden	1	(omitted)			
	Spain	1	(omitted)			
	Italy	1	(omitted)			
	France	1	(omitted)			
	Denmark	1	(omitted)			
	Greece	1	(omitted)			
	Switzerland	1	(omitted)			
	Belgium	1	(omitted)			
	Czechia	1	(omitted)			

Poland	1 (omitted)
--------	-------------

```
168 outreg2 using results2.xls, excel replace dec(3) alpha(0.01, 0.05, 0.10) symbol(***,
> **, *) ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

169

```
170 xtlogit phinact age yedu i.gender i.thinc i.mst i.sphus i.cjs c.bmi ib1.changed##c.e
> urod i.country ///
> Mundlak_Sph_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Gi_* Mundlak_Mar_* //
> /
> , or baselevels vce(robust)
note: Mundlak_Sph_sphus_Poor omitted because of collinearity.
note: Mundlak_C_changed_Newly omitted because of collinearity.
note: Mundlak_Oth_cjs_Other omitted because of collinearity.
note: Mundlak_Oth_thinc_Above omitted because of collinearity.
note: Mundlak_Gi_3 omitted because of collinearity.
note: Mundlak_Mar_mst_Lost omitted because of collinearity.
```

Fitting comparison model:

```
Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -33060.574
Iteration 2: Log pseudolikelihood = -30036.746
Iteration 3: Log pseudolikelihood = -29743.012
Iteration 4: Log pseudolikelihood = -29741.169
Iteration 5: Log pseudolikelihood = -29741.169
```

Fitting full model:

```
tau = 0.0 Log pseudolikelihood = -29741.169
tau = 0.1 Log pseudolikelihood = -29577.101
tau = 0.2 Log pseudolikelihood = -29423.123
tau = 0.3 Log pseudolikelihood = -29291.57
tau = 0.4 Log pseudolikelihood = -29203.444
tau = 0.5 Log pseudolikelihood = -29195.783
tau = 0.6 Log pseudolikelihood = -29337.978
```

```
Iteration 0: Log pseudolikelihood = -29195.783
Iteration 1: Log pseudolikelihood = -28628.647
Iteration 2: Log pseudolikelihood = -28614.915
Iteration 3: Log pseudolikelihood = -28614.881
Iteration 4: Log pseudolikelihood = -28614.881
```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Random effects $u_i \sim \text{Gaussian}$ Integration method: **mvaghermite**Log pseudolikelihood = **-28614.881**

Number of obs = **110,045**
Number of groups = **48,341**

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration pts. = **12**

Wald chi2(56) = **6953.21**
Prob > chi2 = **0.0000**

(Std. err. adjusted for **48,341** clust

> ers in id)

		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]							
		age	1.188615	.007333	28.01	0.000	1.174329
>	1.203075	yedu	.9535701	.0160562	-2.82	0.005	.9226142
>	.9855647						
		gender					
		Male	1 (base)				
		Female	1.159138	.0438393	3.90	0.000	1.076322
>	1.248327						
		thinc					
		No income	1 (base)				
		Below country average income	1.089443	.1544691	0.60	0.546	.8251159
>	1.438448						
		Above country average income	1.070404	.1579783	0.46	0.645	.8015327
>	1.429469						
		mst					
		Officially with partner	1 (base)				
		Away from partner	.9888254	.344199	-0.03	0.974	.4998341
>	1.9562						
		Never married	.946804	.4490791	-0.12	0.908	.3737027
>	2.398799						
		Lost partner	1.13978	.1833802	0.81	0.416	.8315158
>	1.562326						
		sphus					
		Excellent	1 (base)				
		Very good	.8356499	.1066159	-1.41	0.159	.6507653
>	1.073061						
		Good	.9654935	.1213172	-0.28	0.780	.754733
>	1.235109						
		Fair	1.616925	.2094518	3.71	0.000	1.254376
>	2.084261						
		Poor	4.711824	.659557	11.07	0.000	3.581283
>	6.199255						
		cjs					
		Retired	1 (base)				
		Employed or self-employed	1.695272	.177925	5.03	0.000	1.380075
>	2.082458						
		Unemployed	1.625568	.3082588	2.56	0.010	1.120961
>	2.357326						
		Permanently sick	2.214297	.2797972	6.29	0.000	1.728537
>	2.836568						
		Homemaker	1.130684	.1221156	1.14	0.255	.9149773
>	1.397244						
		Other	1.622619	.2358802	3.33	0.001	1.22033
>	2.157524						
		bmi	.9922487	.0084417	-0.91	0.360	.9758405
>	1.008933						
		changed					
		No longer officially together	1.148869	.25942	0.61	0.539	.7380151
>	1.788446						
		All other situations	1 (base)				
		Newly officially together	.9064837	.4356901	-0.20	0.838	.3533788
>	2.325302						
		eurod	1.095156	.0110281	9.03	0.000	1.073753
>	1.116986						

No longer officially together	changed#c.eurod					
> 1.030444		.9560098	.0365714	-1.18	0.240	.8869524
> 1.325256	Newly officially together	1.04394	.1270881	0.35	0.724	.8223396
	country					
	Austria	1	(base)			
> .7633415	Germany	.6389733	.057979	-4.94	0.000	.5348679
> .498293	Sweden	.410042	.0407808	-8.96	0.000	.3374209
> 2.052025	Spain	1.750918	.1417616	6.92	0.000	1.493995
> 6.042848	Italy	5.172617	.4103777	20.71	0.000	4.427707
> 1.457155	France	1.223944	.1089126	2.27	0.023	1.028057
> .8130463	Denmark	.6668018	.0674623	-4.01	0.000	.5468625
> 1.078219	Greece	.9099994	.0787545	-1.09	0.276	.7680247
> .6944123	Switzerland	.5592371	.0617718	-5.26	0.000	.4503752
> 2.064072	Belgium	1.75609	.1447823	6.83	0.000	1.494062
> 1.289818	Czechia	1.093102	.0922918	1.05	0.292	.9263878
> 1.737058	Poland	1.433114	.1406402	3.67	0.000	1.182353
Mundlak_Sph_sphus_Excellent		.0890035	.0175344	-12.28	0.000	.0604942
> .1309484	Mundlak_Sph_sphus_Very	.1013939	.0136723	-16.97	0.000	.0778454
> .132066	Mundlak_Sph_sphus_Good	.1437757	.0144941	-19.24	0.000	.1179982
> .1751845	Mundlak_Sph_sphus_Fair	.3190805	.0282952	-12.88	0.000	.2681749
> .3796491	Mundlak_Sph_sphus_Poor	1	(omitted)			
> .2575786	Mundlak_C_changed_No	.0302029	.0330294	-3.20	0.001	.0035415
> .9831328	Mundlak_C_changed_All	.1362996	.137407	-1.98	0.048	.0188963
> 1.270214	Mundlak_C_changed_Newly	1	(omitted)			
> 1.035046	Mundlak_Oth_cjs_Retired	.8486315	.1746287	-0.80	0.425	.5669719
> 1.823361	Mundlak_Oth_cjs_Employed	.6560143	.1526327	-1.81	0.070	.4157834
> 3.060621	Mundlak_Oth_cjs_Unemployed	.9894193	.3086029	-0.03	0.973	.5368934
> 1.369901	Mundlak_Oth_cjs_Permanently	1.865298	.471279	2.47	0.014	1.136807
> 8.819057	Mundlak_Oth_cjs_Homemaker	.8899645	.1958467	-0.53	0.596	.5781709
> 1.464507	Mundlak_Oth_cjs_Other	1	(omitted)			
> .9267069	Mundlak_Oth_thinc_No	5.798715	1.24047	8.22	0.000	3.812777
> 1.0756	Mundlak_Oth_thinc_Below	1.273219	.0909267	3.38	0.001	1.106916
> 1.085851	Mundlak_Oth_thinc_Above	1	(omitted)			
> 1.320833	Mundlak_OthC_age	.9149927	.0059388	-13.69	0.000	.9034266
> 2.434191	Mundlak_OthC_yedu	1.039514	.0180994	2.23	0.026	1.004638
	Mundlak_OthC_bmi	1.066315	.0098773	6.93	0.000	1.04713
	Mundlak_OthC_eurod	.797309	.2053414	-0.88	0.379	.4812886
	Mundlak_Gi_1	1.436781	.3864785	1.35	0.178	.8480603
	Mundlak_Gi_2	1.278069	.3294809	0.95	0.341	.7711134

>	2.118316	Mundlak_Gi_3	1	(omitted)				
		Mundlak_Mar_mst_Officially	1.044456	.1762984	0.26	0.797	.7502605	
>	1.454013	Mundlak_Mar_mst_Away	1.010859	.3541812	0.03	0.975	.5086854	
>	2.008776	Mundlak_Mar_mst_Never	1.356821	.6558235	0.63	0.528	.5261282	
>	3.499076	Mundlak_Mar_mst_Lost	1	(omitted)				
		_cons	.0002649	.0002897	-7.53	0.000	.000031	
>	.00226							
>	1.092381	/lnsig2u	1.015944	.0389989			.9395078	
>	1.726662	sigma_u	1.661918	.0324065			1.5996	
>	.4754031	rho	.4563853	.0096755			.4374938	

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

171 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
 > ctitle(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
 dir : seeout

172 asdoc, text(xtologitLim) save(report.doc) append
 Click to Open File: [report.doc](#)

173 asdoc test Mundlak_Sph_sphus_Excellent Mundlak_Sph_sphus_Very Mundlak_Sph_sphus_Good
 > Mundlak_Sph_sphus_Fair Mundlak_Sph_sphus_Poor Mundlak_C_changed No Mundlak_C_change
 > d All Mundlak_C_changed Newly Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs_Employed Mundl
 > ak Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Homemaker Mundlak_
 > Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below Mundlak_Oth_thinc_Above M
 > undlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak_OthC_eurod Mundlak_Gi_1 M
 > undlak_Gi_2 Mundlak_Gi_3 Mundlak_Mar_mst_Officially Mundlak_Mar_mst_Away Mundlak_Mar
 > _mst_Never Mundlak_Mar_mst_Lost, save(report.doc) append

```
( 1) [phinact]Mundlak_Sph_sphus_Excellent = 0
( 2) [phinact]Mundlak_Sph_sphus_Very = 0
( 3) [phinact]Mundlak_Sph_sphus_Good = 0
( 4) [phinact]Mundlak_Sph_sphus_Fair = 0
( 5) [phinact]o.Mundlak_Sph_sphus_Poor = 0
( 6) [phinact]Mundlak_C_changed_No = 0
( 7) [phinact]Mundlak_C_changed_All = 0
( 8) [phinact]o.Mundlak_C_changed_Newly = 0
( 9) [phinact]Mundlak_Oth_cjs_Retired = 0
(10) [phinact]Mundlak_Oth_cjs_Employed = 0
(11) [phinact]Mundlak_Oth_cjs_Unemployed = 0
(12) [phinact]Mundlak_Oth_cjs_Permanently = 0
(13) [phinact]Mundlak_Oth_cjs_Homemaker = 0
(14) [phinact]o.Mundlak_Oth_cjs_Other = 0
(15) [phinact]Mundlak_Oth_thinc_No = 0
(16) [phinact]Mundlak_Oth_thinc_Below = 0
(17) [phinact]o.Mundlak_Oth_thinc_Above = 0
(18) [phinact]Mundlak_OthC_age = 0
(19) [phinact]Mundlak_OthC_yedu = 0
(20) [phinact]Mundlak_OthC_bmi = 0
(21) [phinact]Mundlak_OthC_eurod = 0
(22) [phinact]Mundlak_Gi_1 = 0
(23) [phinact]Mundlak_Gi_2 = 0
(24) [phinact]o.Mundlak_Gi_3 = 0
(25) [phinact]Mundlak_Mar_mst_Officially = 0
(26) [phinact]Mundlak_Mar_mst_Away = 0
(27) [phinact]Mundlak_Mar_mst_Never = 0
(28) [phinact]o.Mundlak_Mar_mst_Lost = 0
Constraint 5 dropped
```

Constraint 8 dropped
 Constraint 14 dropped
 Constraint 17 dropped
 Constraint 24 dropped
 Constraint 28 dropped

chi2(22) = **851.00**
 Prob > chi2 = **0.0000**

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174

175 logit phinact age yedu i.gender i.thinc i.mst i.sphus i.cjs c.bmi ib1.changed##c.eur
 > od i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

Iteration 0: Log pseudolikelihood = **-1.011e+08**
 Iteration 1: Log pseudolikelihood = **-84194755**
 Iteration 2: Log pseudolikelihood = **-78010964**
 Iteration 3: Log pseudolikelihood = **-77790841**
 Iteration 4: Log pseudolikelihood = **-77789818**
 Iteration 5: Log pseudolikelihood = **-77789818**

Logistic regression

Number of obs = **110,007**
 Wald chi2(37) = **5584.34**
 Prob > chi2 = **0.0000**
 Pseudo R2 = **0.2304**

Log pseudolikelihood = **-77789818**

(Std. err. adjusted for **48,310** clust

> ers in id)

		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]							
		age	1.059455	.0034042	17.97	0.000	1.052804
>	1.066148	yedu	.9836221	.0055529	-2.93	0.003	.9727986
>	.9945661						
		gender					
		Male	1	(base)			
		Female	1.051987	.0485508	1.10	0.272	.9610058
>	1.151581						
		thinc					
		No income	1	(base)			
		Below country average income	.4782865	.073734	-4.78	0.000	.353561
>	.6470112						
		Above country average income	.4014244	.0627918	-5.84	0.000	.2954315
>	.5454447						
		mst					
		Officially with partner	1	(base)			
		Away from partner	1.212664	.1078036	2.17	0.030	1.018757
>	1.44348						
		Never married	1.228286	.1097548	2.30	0.021	1.030954
>	1.463388						
		Lost partner	1.123588	.0626049	2.09	0.036	1.007347
>	1.253242						
		sphus					
		Excellent	1	(base)			
		Very good	1.351803	.1937132	2.10	0.035	1.02079
>	1.790155						
		Good	1.47709	.1929837	2.99	0.003	1.143396
>	1.908172						
		Fair	3.193801	.4175029	8.88	0.000	2.471931
>	4.126477						
		Poor	12.00211	1.632651	18.27	0.000	9.193237
>	15.6692						

	cjs					
	Retired	1	(base)			
>	Employed or self-employed	1.113648	.0935604	1.28	0.200	.9445745
>	1.312984					
>	Unemployed	1.796336	.3050601	3.45	0.001	1.287756
>	2.505773					
>	Permanently sick	3.063933	.3396972	10.10	0.000	2.465511
>	3.807604					
>	Homemaker	1.180292	.0819923	2.39	0.017	1.03005
>	1.352448					
>	Other	1.452461	.1719625	3.15	0.002	1.151669
>	1.831815					
>	bmi	1.037251	.004649	8.16	0.000	1.028179
>	1.046403					
	changed					
No longer officially together		1.097341	.2631759	0.39	0.699	.6857998
>	1.755843					
	All other situations	1	(base)			
Newly officially together		1.645456	.6961027	1.18	0.239	.7181076
>	3.770363					
	eurod	1.07834	.009003	9.03	0.000	1.060838
>	1.096131					
	changed#c.eurod					
No longer officially together		.9704425	.0390438	-0.75	0.456	.8968574
>	1.050065					
Newly officially together		.9680485	.0925474	-0.34	0.734	.8026395
>	1.167545					
	country					
	Austria	1	(base)			
>	Germany	.8509982	.0739723	-1.86	0.063	.717693
>	1.009064					
>	Sweden	.4867902	.0505345	-6.93	0.000	.3971707
>	.5966319					
>	Spain	1.755929	.165207	5.98	0.000	1.46023
>	2.111506					
>	Italy	3.912001	.2996614	17.81	0.000	3.366639
>	4.545708					
>	France	1.212807	.0991285	2.36	0.018	1.033282
>	1.423523					
>	Denmark	.7756956	.0725598	-2.72	0.007	.6457562
>	.9317816					
>	Greece	1.027192	.0816644	0.34	0.736	.8789801
>	1.200396					
>	Switzerland	.6256717	.0667967	-4.39	0.000	.5075423
>	.7712955					
>	Belgium	1.495425	.1196363	5.03	0.000	1.278402
>	1.749291					
>	Czechia	1.12234	.1194206	1.08	0.278	.9110741
>	1.382595					
>	Poland	1.423411	.1295455	3.88	0.000	1.190863
>	1.701369					
	Qyear					
	2015	1	(base)			
>	2017	.8009518	.0377903	-4.70	0.000	.7302058
>	.8785521					
>	2020	1.028733	.0503088	0.58	0.562	.9347079
>	1.132217					
>	2022	1.058167	.0479857	1.25	0.212	.9681758
>	1.156524					
>	_cons	.0003829	.0001392	-21.64	0.000	.0001877
>	.0007809					

Note: **_cons** estimates baseline odds.

```

176 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout

```

```
177
```

```
178
```

```

179 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ib1.schanged
> c.eurod i.country

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -40209.792
Iteration 1: Log likelihood = -33347.304
Iteration 2: Log likelihood = -30411.484
Iteration 3: Log likelihood = -30168.81
Iteration 4: Log likelihood = -30167.543
Iteration 5: Log likelihood = -30167.543

```

Fitting full model:

```

tau = 0.0 Log likelihood = -30167.543
tau = 0.1 Log likelihood = -30008.398
tau = 0.2 Log likelihood = -29859.539
tau = 0.3 Log likelihood = -29733.391
tau = 0.4 Log likelihood = -29651.002
tau = 0.5 Log likelihood = -29649.359
tau = 0.6 Log likelihood = -29797.475

```

```

Iteration 0: Log likelihood = -29649.359
Iteration 1: Log likelihood = -29122.744
Iteration 2: Log likelihood = -29119.566
Iteration 3: Log likelihood = -29119.559
Iteration 4: Log likelihood = -29119.559

```

Random-effects logistic regression
Group variable: **id**

Number of obs = **110,045**
Number of groups = **48,341**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.3**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-29119.559**

Wald chi2(30) = **7810.69**
Prob > chi2 = **0.0000**

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.0917619	.0022403	40.96	0.000	.087371
> .0961528	yedu	-.0181178	.0040441	-4.48	0.000	-.0260441
> -.0101914	gender					
	Female	.1165455	.0352111	3.31	0.001	.0475331
> .1855579	thinc					
	Below country average income	-.5704357	.0950934	-6.00	0.000	-.7568153
> -.3840561	Above country average income	-.7157181	.0981184	-7.29	0.000	-.9080267
> -.5234095	single					
	Yes	.1375728	.0373086	3.69	0.000	.0644492
> .2106964						

	sphus					
> .1560854	Very good	-.0367122	.0983679	-0.37	0.709	-.2295098
> .5291235	Good	.3506882	.0910401	3.85	0.000	.1722529
> 1.508251	Fair	1.328479	.0917221	14.48	0.000	1.148707
> 3.248709	Poor	3.05736	.097629	31.32	0.000	2.86601
	cjs					
> .416694	Employed or self-employed	.2987743	.0601642	4.97	0.000	.1808546
> .841769	Unemployed	.6113516	.1175621	5.20	0.000	.3809341
> 1.594217	Permanently sick	1.444181	.0765501	18.87	0.000	1.294146
> .2975041	Homemaker	.1935064	.053061	3.65	0.000	.0895087
> .7986266	Other	.6138352	.0942831	6.51	0.000	.4290439
> .0548553	bmi	.0486491	.0031665	15.36	0.000	.0424428
> .336454	schanged Hooked up	-.1197227	.2327475	-0.51	0.607	-.5758995
> -.0658363	Back single	-.2195829	.0784436	-2.80	0.005	-.3733295
> .1280908	eurod	.1157354	.0063039	18.36	0.000	.1033801
> -.1633558	country Germany	-.3368096	.0884985	-3.81	0.000	-.5102633
> -.7123111	Sweden	-.9046483	.0981331	-9.22	0.000	-1.096986
> .7535724	Spain	.5964797	.0801508	7.44	0.000	.4393869
> 1.816442	Italy	1.663995	.0777802	21.39	0.000	1.511549
> .4085393	France	.2395919	.0861992	2.78	0.005	.0706445
> -.1955677	Denmark	-.389312	.098851	-3.94	0.000	-.5830563
> .1045782	Greece	-.0578875	.0828922	-0.70	0.485	-.2203533
> -.4639048	Switzerland	-.6735124	.1069446	-6.30	0.000	-.88312
> .7234424	Belgium	.564652	.081017	6.97	0.000	.4058615
> .323396	Czechia	.1598944	.0834207	1.92	0.055	-.0036071
> .6616726	Poland	.4737736	.0958686	4.94	0.000	.2858745
> -11.31643	_cons	-11.82667	.2603326	-45.43	0.000	-12.33691
> .9957544	/lnsig2u	.9199512	.0386758			.8441481
> 1.645225	sigma_u	1.584035	.0306319			1.525121
> .4513808	rho	.4326871	.0094937			.414183

LR test of rho=0: chibar2(01) = 2095.97

Prob >= chibar2 = 0.000

180 estimates store RE

181 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ib1.schanged
 > c.eurod i.country, fe

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **19.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -5684.9537

Iteration 1: Log likelihood = -5284.3129

Iteration 2: Log likelihood = -5282.3016

Iteration 3: Log likelihood = -5282.301

Iteration 4: Log likelihood = -5282.301

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = 17,262

Number of groups = 5,962

Obs per group:

min = 2

avg = 2.9

max = 4

Log likelihood = -5282.301

LR chi2(18) = 2033.13

Prob > chi2 = 0.0000

	phinact	Coefficient	Std. err.	z	P> z	[95% conf.
> interval]						
	age	.1736827	.0061113	28.42	0.000	.1617047
> .1856606	yedu	-.041587	.0195352	-2.13	0.033	-.0798752
> -.0032989						
	gender					
	Female	0	(omitted)			
	thinc					
Below country average income		.1518173	.1266738	1.20	0.231	-.0964587
> .4000933						
Above country average income		.1023607	.1344671	0.76	0.447	-.16119
> .3659115						
	single					
	Yes	.0237966	.142113	0.17	0.867	-.2547397
> .3023329						
	sphus					
	Very good	-.2041543	.1434483	-1.42	0.155	-.4853079
> .0769992						
	Good	-.0334229	.1388319	-0.24	0.810	-.3055284
> .2386826						
	Fair	.4375895	.1418472	3.08	0.002	.1595741
> .715605						
	Poor	1.375489	.150051	9.17	0.000	1.081394

> 1.669583						
	cjs					
Employed or self-employed		.5093894	.1030404	4.94	0.000	.3074339
> .7113448						
Unemployed		.5618251	.1761819	3.19	0.001	.216515
> .9071353						
Permanently sick		.761857	.1259498	6.05	0.000	.5149998
> 1.008714						
Homemaker		.1334421	.0985912	1.35	0.176	-.0597932
> .3266773						
Other		.4734759	.1416124	3.34	0.001	.1959207
> .7510312						
bmi		-.0029733	.0075599	-0.39	0.694	-.0177905
> .0118438						
schanged						
Hooked up		-.2134335	.2902511	-0.74	0.462	-.7823153
> .3554483						
Back single		-.1278368	.1386576	-0.92	0.357	-.3996007
> .1439271						
eurod		.0762879	.0094177	8.10	0.000	.0578295
> .0947463						
country						
Germany		0	(omitted)			
Sweden		0	(omitted)			
Spain		0	(omitted)			
Italy		0	(omitted)			
France		0	(omitted)			
Denmark		0	(omitted)			
Greece		0	(omitted)			
Switzerland		0	(omitted)			
Belgium		0	(omitted)			
Czechia		0	(omitted)			
Poland		0	(omitted)			

182 estimates store FE

183 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients		(b-B)	sqrt(diag(V_b-V_B))
	(b)	(B)	Difference	Std. err.
	FE	RE		
age	.1736827	.0917619	.0819207	.0056859
yedu	-.041587	-.0181178	-.0234693	.019112
thinc				
1	.1518173	-.5704357	.722253	.0836869
2	.1023607	-.7157181	.8180788	.0919466
1.single	.0237966	.1375728	-.1137762	.1371283
sphus				
2	-.2041543	-.0367122	-.1674421	.1044087
3	-.0334229	.3506882	-.3841111	.1048141
4	.4375895	1.328479	-.8908899	.1082021
5	1.375489	3.05736	-1.681871	.1139469
cjs				
2	.5093894	.2987743	.2106151	.0836516
3	.5618251	.6113516	-.0495264	.131222
4	.761857	1.444181	-.6823245	.1000172
5	.1334421	.1935064	-.0600643	.0830949
6	.4734759	.6138352	-.1403593	.1056635
bmi	-.0029733	.0486491	-.0516224	.0068648
schanged				
0	-.2134335	-.1197227	-.0937108	.1734195
2	-.1278368	-.2195829	.0917461	.1143352
eurod	.0762879	.1157354	-.0394475	.0069968

b = Consistent under H0 and Ha; obtained from **xtlogit**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(18) = (b-B)'[(V_b-V_B)^(-1)](b-B)
 = **1179.39**
 Prob > chi2 = **0.0000**
 Click to Open File: [report.doc](#)

184

185

186 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ibl.schanged#

> #c.eurod i.country, fe or baselevels

note: multiple positive outcomes within groups encountered.

note: 42,379 groups (92,783 obs) omitted because of all positive or
 all negative outcomes.

note: **2.gender** omitted because of no within-group variance.

note: **12.country** omitted because of no within-group variance.

note: **13.country** omitted because of no within-group variance.

note: **15.country** omitted because of no within-group variance.

note: **16.country** omitted because of no within-group variance.

note: **17.country** omitted because of no within-group variance.

note: **18.country** omitted because of no within-group variance.

note: **19.country** omitted because of no within-group variance.

note: **20.country** omitted because of no within-group variance.

note: **23.country** omitted because of no within-group variance.

note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-5682.7646**

Iteration 1: Log likelihood = **-5282.4307**

Iteration 2: Log likelihood = **-5280.405**

Iteration 3: Log likelihood = **-5280.4044**

Iteration 4: Log likelihood = **-5280.4044**

Conditional fixed-effects logistic regression
 Group variable: **id**

Number of obs = **17,262**
 Number of groups = **5,962**

Obs per group:
 min = **2**
 avg = **2.9**
 max = **4**

Log likelihood = **-5280.4044**

LR chi2(20) = **2036.93**
 Prob > chi2 = **0.0000**

	phinact	Odds ratio	Std. err.	z	P> z	[95% conf.]
> interval]						
	age	1.189626	.0072737	28.40	0.000	1.175455
> 1.203968	yedu	.9592752	.0187274	-2.13	0.033	.9232636
> .9966915						
	gender					
	Male	1	(base)			
	Female	1	(omitted)			
	thinc					
	No income	1	(base)			
Below country average income		1.159993	.1469164	1.17	0.241	.9049996
> 1.486834						
Above country average income		1.105394	.1486059	0.75	0.456	.8493446
> 1.438633						
	single					
	No	1	(base)			
	Yes	1.022438	.1457425	0.16	0.876	.7732201

> 1.351981						
	sphus					
	Excellent	1 (base)				
> 1.086753	Very good	.8201882	.1177644	-1.38	0.167	.6190079
> 1.275963	Good	.971781	.1350236	-0.21	0.837	.740114
> 2.054956	Fair	1.555829	.220873	3.11	0.002	1.177934
> 5.323234	Poor	3.966017	.5955585	9.17	0.000	2.954837
	cjs					
	Retired	1 (base)				
> 2.045722	Employed or self-employed	1.671461	.1723108	4.98	0.000	1.365671
> 2.466657	Unemployed	1.745641	.3079349	3.16	0.002	1.235382
> 2.737695	Permanently sick	2.138735	.2694218	6.03	0.000	1.670818
> 1.38247	Homemaker	1.139352	.112433	1.32	0.186	.9389882
> 2.12077	Other	1.606685	.2275684	3.35	0.001	1.217217
> 1.012147	bmi	.9972502	.0075442	-0.36	0.716	.9825729
> 1.182089	schanged Hooked up	.4670907	.22128	-1.61	0.108	.1845662
> 1.57344	Unchanged Back single	1 (base) 1.052301	.2159861	0.25	0.804	.7037689
> 1.101434	eurod	1.080849	.0104042	8.08	0.000	1.060648
> 1.479475	schanged#c.eurod Hooked up	1.182463	.135194	1.47	0.143	.9450776
> 1.026777	Back single	.9616233	.0321645	-1.17	0.242	.9006039
	country					
	Austria	1 (base)				
	Germany	1 (omitted)				
	Sweden	1 (omitted)				
	Spain	1 (omitted)				
	Italy	1 (omitted)				
	France	1 (omitted)				
	Denmark	1 (omitted)				
	Greece	1 (omitted)				
	Switzerland	1 (omitted)				
	Belgium	1 (omitted)				
	Czechia	1 (omitted)				
	Poland	1 (omitted)				

```
187 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

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```
189 xtlogit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ibl.schanged#
> #c.eurod i.country ///
> Mundlak_Sph_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_sGi_* Mundlak_Sin_*
> ///
> , or baselevels vce(robust)
note: Mundlak_Sph_sphus_Poor omitted because of collinearity.
note: Mundlak_sC_schanged_Back omitted because of collinearity.
note: Mundlak_Oth_cjs_Other omitted because of collinearity.
note: Mundlak_Oth_thinc_Above omitted because of collinearity.
note: Mundlak_sGi_3 omitted because of collinearity.
note: Mundlak_Sin_single_Yes omitted because of collinearity.
```

Fitting comparison model:

```
Iteration 0: Log pseudolikelihood = -40209.792
Iteration 1: Log pseudolikelihood = -33068.857
Iteration 2: Log pseudolikelihood = -30049.319
Iteration 3: Log pseudolikelihood = -29755.169
Iteration 4: Log pseudolikelihood = -29753.329
Iteration 5: Log pseudolikelihood = -29753.328
```

Fitting full model:

```
tau = 0.0 Log pseudolikelihood = -29753.328
tau = 0.1 Log pseudolikelihood = -29589.191
tau = 0.2 Log pseudolikelihood = -29435.101
tau = 0.3 Log pseudolikelihood = -29303.399
tau = 0.4 Log pseudolikelihood = -29215.085
tau = 0.5 Log pseudolikelihood = -29207.188
tau = 0.6 Log pseudolikelihood = -29349.073
```

```
Iteration 0: Log pseudolikelihood = -29207.188
Iteration 1: Log pseudolikelihood = -28639.683
Iteration 2: Log pseudolikelihood = -28625.477
Iteration 3: Log pseudolikelihood = -28625.441
Iteration 4: Log pseudolikelihood = -28625.441
```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Number of obs = 110,045
Number of groups = 48,341

Random effects u_i ~ Gaussian

Obs per group:
min = 1
avg = 2.3
max = 4

Integration method: **mvaghermite**

Integration pts. = 12

Log pseudolikelihood = -28625.441

Wald chi2(52) = 6941.96
Prob > chi2 = 0.0000

(Std. err. adjusted for 48,341 clusters)

> rs in **id**)

	phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
> interval]						
	age	1.190097	.0074097	27.95	0.000	1.175663
> 1.204709	yedu	.9535509	.0160801	-2.82	0.005	.9225496
> .9855939						

	gender					
	Male	1	(base)			
	Female	1.148887	.0434805	3.67	0.000	1.066751
>	1.237347					
	thinc					
	No income	1	(base)			
	Below country average income	1.084078	.1537986	0.57	0.569	.8209193
>	1.431598					
	Above country average income	1.064795	.1574844	0.42	0.671	.7968422
>	1.422852					
	single					
	No	1	(base)			
	Yes	1.092158	.155997	0.62	0.537	.8254774
>	1.444993					
	sphus					
	Excellent	1	(base)			
	Very good	.8377234	.1067954	-1.39	0.165	.6525098
>	1.07551					
	Good	.9656618	.1212915	-0.28	0.781	.7549363
>	1.235207					
	Fair	1.616233	.2092914	3.71	0.000	1.253947
>	2.08319					
	Poor	4.708454	.6589117	11.07	0.000	3.57898
>	6.194374					
	cjs					
	Retired	1	(base)			
	Employed or self-employed	1.703181	.1789031	5.07	0.000	1.386277
>	2.092529					
	Unemployed	1.630076	.3097113	2.57	0.010	1.123263
>	2.365563					
	Permanently sick	2.218782	.2807348	6.30	0.000	1.73147
>	2.843245					
	Homemaker	1.130848	.1221764	1.14	0.255	.9150415
>	1.397551					
	Other	1.623955	.2360288	3.34	0.001	1.221402
>	2.159182					
	bmi	.9923779	.0084427	-0.90	0.368	.9759677
>	1.009064					
	schanged					
	Hooked up	.4326827	.227825	-1.59	0.112	.154162
>	1.2144					
	Unchanged	1	(base)			
	Back single	1.027218	.2065957	0.13	0.894	.6925759
>	1.523553					
	eurod	1.095985	.0110696	9.07	0.000	1.074502
>	1.117897					
	schanged#c.eurod					
	Hooked up	1.238291	.1351873	1.96	0.050	.999758
>	1.533736					
	Back single	.9603416	.0330998	-1.17	0.240	.8976099
>	1.027457					
	country					
	Austria	1	(base)			
	Germany	.6425898	.0583202	-4.87	0.000	.5378739
>	.7676922					
	Sweden	.4104047	.0408456	-8.95	0.000	.3376731
>	.4988021					
	Spain	1.760354	.1423294	6.99	0.000	1.502374
>	2.062633					
	Italy	5.202455	.411893	20.83	0.000	4.454678
>	6.075757					
	France	1.231873	.1096248	2.34	0.019	1.034707
>	1.466611					

> .8185667	Denmark	.6711524	.0679926	-3.94	0.000	.5502857
> 1.076887	Greece	.9091164	.078555	-1.10	0.270	.767483
> .6956975	Switzerland	.5603593	.0618513	-5.25	0.000	.4513493
> 2.078114	Belgium	1.767823	.1458589	6.91	0.000	1.503862
> 1.285493	Czechia	1.089834	.091813	1.02	0.307	.923955
> 1.743141	Poland	1.438956	.1407933	3.72	0.000	1.187853
Mundlak_Sph_sphus_Excellent		.0886598	.0174753	-12.29	0.000	.0602492
> .1304676	Mundlak_Sph_sphus_Very	.1007287	.0135828	-17.02	0.000	.0773344
> .1312001	Mundlak_Sph_sphus_Good	.1431355	.0144284	-19.28	0.000	.1174747
> .1744016	Mundlak_Sph_sphus_Fair	.3186253	.0282412	-12.90	0.000	.2678147
> .3790758	Mundlak_Sph_sphus_Poor	1	(omitted)			
Mundlak_sC_schanged_Hooked		5.382566	5.16367	1.75	0.079	.821115
> 35.28375	Mundlak_sC_schanged_Unchanged	2.099792	.7777877	2.00	0.045	1.015971
> 4.339817	Mundlak_sC_schanged_Back	1	(omitted)			
Mundlak_Oth_cjs_Retired		.8413676	.172941	-0.84	0.401	.5623716
> 1.258775	Mundlak_Oth_cjs_Employed	.6493717	.1508833	-1.86	0.063	.4118265
> 1.023935	Mundlak_Oth_cjs_Unemployed	.9903497	.3083724	-0.03	0.975	.5379522
> 1.823196	Mundlak_Oth_cjs_Permanently	1.873852	.473637	2.48	0.013	1.141785
> 3.075289	Mundlak_Oth_cjs_Homemaker	.8851844	.1947926	-0.55	0.579	.5750682
> 1.362536	Mundlak_Oth_cjs_Other	1	(omitted)			
Mundlak_Oth_thinc_No		5.665016	1.215417	8.08	0.000	3.720297
> 8.626302	Mundlak_Oth_thinc_Below	1.258836	.0913251	3.17	0.002	1.091985
> 1.45118	Mundlak_Oth_thinc_Above	1	(omitted)			
Mundlak_OthC_age		.9136553	.005965	-13.83	0.000	.9020386
> .9254215	Mundlak_OthC_yedu	1.03928	.0181173	2.21	0.027	1.00437
> 1.075403	Mundlak_OthC_bmi	1.066103	.0098793	6.91	0.000	1.046915
> 1.085643	Mundlak_OthC_eurod	1.091552	.0766987	1.25	0.213	.9511177
> 1.252723	Mundlak_sGi_1	.5917118	.1380153	-2.25	0.024	.3746015
> .934654	Mundlak_sGi_2	.9325597	.0664075	-0.98	0.327	.8110781
> 1.072237	Mundlak_sGi_3	1	(omitted)			
Mundlak_Sin_single_No		1.010687	.1509275	0.07	0.943	.7542326
> 1.354341	Mundlak_Sin_single_Yes	1	(omitted)			
Mundlak_Sin_single_Yes	_Cons	.0000186	.0000102	-19.77	0.000	6.31e-06
> .0000547						
> 1.093952	/lnsig2u	1.017592	.0389599			.9412317
> 1.728019	sigma_u	1.663287	.0324008			1.60098
> .475795	rho	.4567941	.0096673			.4379181

Note: Estimates are transformed only in the first equation to odds ratios.
 Note: _cons estimates baseline odds (conditional on zero random effects).

```
190 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *)
> ctitle(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

```
191 asdoc, text(xtologitLim) save(report.doc) append
Click to Open File: report.doc
```

```
192 asdoc test Mundlak_Sph_sphus_Excellent Mundlak_Sph_sphus_Very Mundlak_Sph_sphus_Good
> Mundlak_Sph_sphus_Fair Mundlak_Sph_sphus_Poor Mundlak_sC_schanged_Hooked Mundlak_sC
> _schanged_Unchanged Mundlak_sC_schanged_Back Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs
> _Employed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Hom
> emaker Mundlak_Oth_cjs_Other Mundlak_Oth_thinc_No Mundlak_Oth_thinc_Below Mundlak_Ot
> h_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_OthC_bmi Mundlak_OthC_eurod
> Mundlak_sGi_1 Mundlak_sGi_2 Mundlak_sGi_3 Mundlak_Sin_single_No Mundlak_Sin_single_
> Yes, save(report.doc) append
```

```
( 1) [phinact]Mundlak_Sph_sphus_Excellent = 0
( 2) [phinact]Mundlak_Sph_sphus_Very = 0
( 3) [phinact]Mundlak_Sph_sphus_Good = 0
( 4) [phinact]Mundlak_Sph_sphus_Fair = 0
( 5) [phinact]o.Mundlak_Sph_sphus_Poor = 0
( 6) [phinact]Mundlak_sC_schanged_Hooked = 0
( 7) [phinact]Mundlak_sC_schanged_Unchanged = 0
( 8) [phinact]o.Mundlak_sC_schanged_Back = 0
( 9) [phinact]Mundlak_Oth_cjs_Retired = 0
(10) [phinact]Mundlak_Oth_cjs_Employed = 0
(11) [phinact]Mundlak_Oth_cjs_Unemployed = 0
(12) [phinact]Mundlak_Oth_cjs_Permanently = 0
(13) [phinact]Mundlak_Oth_cjs_Homemaker = 0
(14) [phinact]o.Mundlak_Oth_cjs_Other = 0
(15) [phinact]Mundlak_Oth_thinc_No = 0
(16) [phinact]Mundlak_Oth_thinc_Below = 0
(17) [phinact]o.Mundlak_Oth_thinc_Above = 0
(18) [phinact]Mundlak_OthC_age = 0
(19) [phinact]Mundlak_OthC_yedu = 0
(20) [phinact]Mundlak_OthC_bmi = 0
(21) [phinact]Mundlak_OthC_eurod = 0
(22) [phinact]Mundlak_sGi_1 = 0
(23) [phinact]Mundlak_sGi_2 = 0
(24) [phinact]o.Mundlak_sGi_3 = 0
(25) [phinact]Mundlak_Sin_single_No = 0
(26) [phinact]o.Mundlak_Sin_single_Yes = 0
Constraint 5 dropped
Constraint 8 dropped
Constraint 14 dropped
Constraint 17 dropped
Constraint 24 dropped
Constraint 26 dropped
```

```
chi2( 20) = 839.69
Prob > chi2 = 0.0000
```

Click to Open File: report.doc

```
193
```

```
194 logit phinact age yedu i.gender i.thinc i.single i.sphus i.cjs c.bmi ib1.schanged##c
> .eurod i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)
```

```
Iteration 0: Log pseudolikelihood = -1.011e+08
Iteration 1: Log pseudolikelihood = -84158818
Iteration 2: Log pseudolikelihood = -77976735
Iteration 3: Log pseudolikelihood = -77756657
Iteration 4: Log pseudolikelihood = -77755617
Iteration 5: Log pseudolikelihood = -77755617
```

Logistic regression

Number of obs = **110,007**Wald chi2(35) = **5558.65**Prob > chi2 = **0.0000**Pseudo R2 = **0.2308**Log pseudolikelihood = **-77755617**(Std. err. adjusted for **48,310** clusters)

> rs in id)

> interval]		phinact	Odds ratio	Robust std. err.	z	P> z	[95% conf.
		age	1.057819	.0033249	17.88	0.000	1.051322
>	1.064356	yedu	.983553	.0055296	-2.95	0.003	.9727747
>	.9944507	gender					
		Male	1	(base)			
		Female	1.033548	.0477688	0.71	0.475	.9440386
>	1.131545	thinc					
		No income	1	(base)			
		Below country average income	.4780525	.0731591	-4.82	0.000	.3541696
>	.6452675	Above country average income	.4076607	.0631077	-5.80	0.000	.300974
>	.5521648	single					
		No	1	(base)			
		Yes	1.207747	.0616203	3.70	0.000	1.092816
>	1.334766	sphus					
		Excellent	1	(base)			
		Very good	1.344607	.1928516	2.06	0.039	1.015105
>	1.781065	Good	1.473733	.1926859	2.97	0.003	1.140583
>	1.904192	Fair	3.189276	.4170661	8.87	0.000	2.468194
>	4.121022	Poor	12.00902	1.632356	18.29	0.000	9.200381
>	15.67506	cjs					
		Retired	1	(base)			
		Employed or self-employed	1.109913	.0935637	1.24	0.216	.9408802
>	1.309314	Unemployed	1.80146	.3030793	3.50	0.000	1.295441
>	2.505137	Permanently sick	3.064604	.3383006	10.15	0.000	2.468372
>	3.804856	Homemaker	1.178312	.0816517	2.37	0.018	1.028669
>	1.349723	Other	1.42721	.1680018	3.02	0.003	1.133157
>	1.79757	bmi	1.037215	.0046247	8.19	0.000	1.02819
>	1.046318	schanged					
		Hooked up	.5719507	.2641308	-1.21	0.226	.2313491
>	1.414	Unchanged	1	(base)			
		Back single	.9248256	.1810067	-0.40	0.690	.6301774
>	1.357241	eurod	1.078442	.0090489	9.00	0.000	1.060851
>	1.096324						

	schanged#c.eurod					
> 1.173187	Hooked up	.98425	.0881821	-0.18	0.859	.8257405
> 1.042323	Back single	.9729173	.0342058	-0.78	0.435	.9081329
	country					
	Austria	1 (base)				
> 1.008263	Germany	.8503317	.0739097	-1.87	0.062	.7171386
> .5991395	Sweden	.4883652	.0509382	-6.87	0.000	.3980718
> 2.092146	Spain	1.740039	.1636047	5.89	0.000	1.447191
> 4.495661	Italy	3.872952	.2946181	17.80	0.000	3.336496
> 1.420141	France	1.209747	.0989694	2.33	0.020	1.030523
> .936591	Denmark	.779601	.0729753	-2.66	0.008	.6489255
> 1.186134	Greece	1.015368	.0805303	0.19	0.848	.8691879
> .7730681	Switzerland	.6270144	.0669886	-4.37	0.000	.5085543
> 1.756224	Belgium	1.500648	.1204133	5.06	0.000	1.282264
> 1.373712	Czechia	1.115594	.1184656	1.03	0.303	.9059752
> 1.667675	Poland	1.397131	.1261785	3.70	0.000	1.170477
	Qyear					
	2015	1 (base)				
> .8905923	2017	.8119667	.0382905	-4.42	0.000	.7402827
> 1.151374	2020	1.045917	.0512625	0.92	0.360	.9501196
> 1.177215	2022	1.076896	.0489386	1.63	0.103	.985126
> .0008593	_cons	.0004273	.0001523	-21.76	0.000	.0002124

Note: **_cons** estimates baseline odds.

195 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
 > ctitle(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
 results2.xls
 dir : seeout

196

197 log close InactivityAndMarriage
 name: **InactivityAndMarriage**
 log: C:\Users\hansg\Documents\working\SHARE data 5\output.smcl
 log type: **smcl**
 closed on: **22 Jun 2026, 15:27:29**